

8. Trigonometric and Hyperbolic Functions

If we solve the equations

$$\begin{aligned} e^{i\theta} &= \cos \theta + i \sin \theta, \\ e^{-i\theta} &= \cos \theta - i \sin \theta \end{aligned}$$

for $\cos \theta$ and $\sin \theta$, we obtain

$$\begin{aligned} \cos \theta &= \frac{e^{i\theta} + e^{-i\theta}}{2}, \\ \sin \theta &= \frac{e^{i\theta} - e^{-i\theta}}{2i}. \end{aligned}$$

This motivates us to extend the definition of $\cos z$ and $\sin z$ to complex numbers z by

$$\begin{aligned} \cos z &= \frac{e^{iz} + e^{-iz}}{2}, & z \in \mathbb{C}, \\ \sin z &= \frac{e^{iz} - e^{-iz}}{2i}, & z \in \mathbb{C}. \end{aligned}$$

This definition agrees with the usual definition when z is real. Evidently, $\cos z$ is an even function,

$$\cos(-z) = \cos z, \quad z \in \mathbb{C},$$

while $\sin z$ is an odd function,

$$\sin(-z) = -\sin z, \quad z \in \mathbb{C}.$$

As functions of a complex variable, $\cos z$ and $\sin z$ are periodic, with period 2π ,

$$\begin{aligned} \cos(z + 2\pi) &= \cos z, & z \in \mathbb{C}, \\ \sin(z + 2\pi) &= \sin z, & z \in \mathbb{C}. \end{aligned}$$

After some algebraic manipulation, one checks (Exercise 1) that the addition formulae for $\cos z$ and $\sin z$ remain valid,

$$\begin{aligned} \cos(z + w) &= \cos z \cos w - \sin z \sin w, & z, w \in \mathbb{C}, \\ \sin(z + w) &= \sin z \cos w + \cos z \sin w, & z, w \in \mathbb{C}. \end{aligned}$$

If we substitute $w = -z$ in the addition formula for cosine, we obtain the familiar identity

$$\cos^2 z + \sin^2 z = 1, \quad z \in \mathbb{C}.$$

We shall see, in fact, that any reasonable identity that holds for analytic functions of a real variable, such as $\cos x$ and $\sin x$, also holds when the functions are extended to be functions of a complex variable. This will be a

special case of the **principle of permanence of functional equations**, proved in Chapter V.

The hyperbolic functions $\cosh x = (e^x + e^{-x})/2$ and $\sinh x = (e^x - e^{-x})/2$ are also extended to the complex plane in the obvious way, by

$$\begin{aligned}\cosh z &= \frac{e^z + e^{-z}}{2}, & z \in \mathbb{C}, \\ \sinh z &= \frac{e^z - e^{-z}}{2}, & z \in \mathbb{C}.\end{aligned}$$

Both $\cosh z$ and $\sinh z$ are periodic, with period $2\pi i$,

$$\begin{aligned}\cosh(z + 2\pi i) &= \cosh z, & z \in \mathbb{C}, \\ \sinh(z + 2\pi i) &= \sinh z, & z \in \mathbb{C}.\end{aligned}$$

Evidently, $\cosh z$ is an even function and $\sinh z$ is an odd function. There are addition formulae for $\cosh z$ and $\sinh z$, derived easily from the addition formulae for $\cos z$ and $\sin z$ (Exercise 1).

When viewed as functions of a complex variable, the trigonometric and the hyperbolic functions exhibit a close relationship. They are obtained from each other by rotating the domain space by $\pi/2$,

$$\begin{aligned}\cosh(iz) &= \cos z, & \cos(iz) &= \cosh z, \\ \sinh(iz) &= i \sin z, & \sin(iz) &= i \sinh z.\end{aligned}$$

If we use these equations and the addition formula

$$\sin(x + iy) = \sin x \cos(iy) + \cos x \sin(iy),$$

we obtain the Cartesian representation for $\sin z$,

$$\sin z = \sin x \cosh y + i \cos x \sinh y, \quad z = x + iy \in \mathbb{C}.$$

Thus

$$|\sin z|^2 = \sin^2 x \cosh^2 y + \cos^2 x \sinh^2 y.$$

Using $\cos^2 x + \sin^2 x = 1$ and $\cosh^2 y = 1 + \sinh^2 y$, we obtain

$$|\sin z|^2 = \sin^2 x + \sinh^2 y.$$

From this formula it is clear where the zeros of $\sin z$ are located; $\sin z = 0$ only when $\sin x = 0$ and $\sinh y = 0$, and this occurs only on the real axis $y = 0$, at the usual zeros $0, \pm\pi, \pm2\pi, \dots$ of $\sin x$. Similarly, the only zeros of $\cos z$ are the usual zeros of $\cos x$ on the real axis (Exercise 2).

Other trigonometric and hyperbolic functions are defined by the usual formulae, such as

$$\tan z = \frac{\sin z}{\cos z}, \quad \tanh z = \frac{\sinh z}{\cosh z}, \quad z \in \mathbb{C}.$$

Thus $\tan z$ and $\tanh z$ are odd functions, and $\tanh(iz) = i \tan z$.

The inverse trigonometric functions are multivalued functions, which can be expressed in terms of the logarithm function. Suppose $w = \sin^{-1} z$, that is,

$$\sin w = \frac{e^{iw} - e^{-iw}}{2i} = z.$$

Then $e^{2iw} - 2ize^{iw} - 1 = 0$. This is a quadratic equation in e^{iw} , which can be solved by the usual quadratic formula. The solutions are given by

$$e^{iw} = iz \pm \sqrt{1 - z^2}.$$

Taking logarithms we obtain

$$\sin^{-1} z = -i \log \left(iz \pm \sqrt{1 - z^2} \right).$$

This identity is to be understood as a set identity, in the sense that w satisfies $\sin w = z$ if and only if w is one of the values of $-i \log \left(iz \pm \sqrt{1 - z^2} \right)$. To obtain a genuine function, we must restrict the domain and specify the branch. One way to do this is to draw two branch cuts, from $-\infty$ to -1 and from $+1$ to $+\infty$ along the real axis, and to specify the branch of $\sqrt{1 - z^2}$ that is positive on the interval $(-1, 1)$. With this branch of $\sqrt{1 - z^2}$, we obtain a continuous branch $-i \operatorname{Log} \left(iz + \sqrt{1 - z^2} \right)$ of $\sin^{-1} z$.

Exercises for I.8

1. Establish the following addition formulae:
 - (a) $\cos(z + w) = \cos z \cos w - \sin z \sin w$,
 - (b) $\sin(z + w) = \sin z \cos w + \cos z \sin w$,
 - (c) $\cosh(z + w) = \cosh z \cosh w + \sinh z \sinh w$,
 - (d) $\sinh(z + w) = \sinh z \cosh w + \cosh z \sinh w$,
2. Show that $|\cos z|^2 = \cos^2 x + \sinh^2 y$, where $z = x + iy$. Find all zeros and periods of $\cos z$.
3. Find all zeros and periods of $\cosh z$ and $\sinh z$.
4. Show that

$$\tan^{-1} z = \frac{1}{2i} \log \left(\frac{1 + iz}{1 - iz} \right),$$

where both sides of the identity are to be interpreted as subsets of the complex plane. In other words, show that $\tan w = z$ if and only if $2iw$ is one of the values of the logarithm featured on the right.

5. Let S denote the two slits along the imaginary axis in the complex plane, one running from i to $+i\infty$, the other from $-i$ to $-i\infty$. Show that $(1 + iz)/(1 - iz)$ lies on the negative real axis $(-\infty, 0]$ if and

only if $z \in S$. Show that the principal branch

$$\operatorname{Tan}^{-1} z = \frac{1}{2i} \operatorname{Log} \left(\frac{1 + iz}{1 - iz} \right)$$

maps the slit plane $\mathbb{C} \setminus S$ one-to-one onto the horizontal strip $\{-\pi < \operatorname{Im} w < \pi\}$.

6. Describe the Riemann surface for $\tan^{-1} z$.
7. Set $w = \cos z$ and $\zeta = e^{iz}$. Show that $\zeta = w \pm \sqrt{w^2 - 1}$. Show that

$$\cos^{-1} w = -i \log \left[w \pm \sqrt{w^2 - 1} \right],$$

where both sides of the identity are to be interpreted as subsets of the complex plane.

8. Show that the vertical strip $|\operatorname{Re}(w)| < \pi/2$ is mapped by the function $z(w) = \sin w$ one-to-one onto the complex z -plane with two slits $(-\infty, -1]$ and $[+1, +\infty)$ on the real axis. Show that the inverse function is the branch of $\sin^{-1} z = -i \operatorname{Log} (iz + \sqrt{1 - z^2})$ obtained by taking the principal value of the square root. *Hint.* First show that the function $1 - z^2$ on the slit plane omits the negative real axis, so that the principal value of the square root is defined and continuous on the slit plane, with argument in the open interval between $-\pi/2$ and $\pi/2$.

II

Analytic Functions

In this chapter we take up the complex differential calculus. After reviewing some basic analysis in Section 1, we introduce complex derivatives and analytic functions in Section 2 and we show that the rules for complex differentiation are the same as the usual rules for differentiation. In Section 3 we characterize analytic functions in terms of the Cauchy-Riemann equations. In Sections 4 and 5 we give several applications of the Cauchy-Riemann equations, to inverses of analytic functions and to harmonic functions. In Section 6 we discuss conformality, which is a direct consequence of complex differentiability. We close in Section 7 with a discussion of fractional linear transformations, which form an important class of analytic functions.

1. Review of Basic Analysis

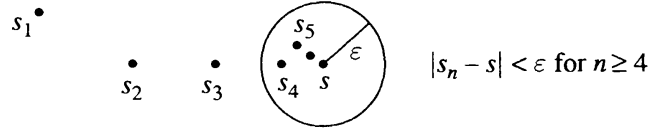
We begin by reviewing the background material in analysis that will (eventually) be called upon, and we say something about the language of formal mathematics. For the most part, we will not phrase our arguments completely formally, though any bilingual person will be able to translate easily to the language of formal mathematics in such a way that our development becomes completely rigorous.

Since the complex derivative is defined as a limit, we require some background material on limits and continuity. To be able to define and work with analytic functions, we also require some basic topological concepts, including open and closed sets, and domains. The confident reader may pass directly to the definitions of complex derivative and analytic function in the next section, and refer back to the material in this section only when needed.

We begin with the notion of a convergent sequence. For this we have two definitions.

Informal Definition. A sequence $\{s_n\}$ **converges to** s if the sequence eventually lies in any disk centered at s .

The language of formal mathematics serves to quantify this statement and make it precise. The “small disk” is traditionally given radius $\varepsilon > 0$. That “ s_n lies in the disk” means that $|s_n - s| < \varepsilon$. That an event “eventually” occurs is translated to the statement that there is $N \geq 1$ such that the event occurs for $n \geq N$. Thus the translation of the definition of convergent sequence to the language of formal mathematics is as follows.



Formal Definition. A sequence of complex numbers $\{s_n\}$ **converges** to s if for any $\varepsilon > 0$, there is an integer $N \geq 1$ such that $|s_n - s| < \varepsilon$ for all $n \geq N$.

If $\{s_n\}$ converges to s , we write $s_n \rightarrow s$, or $\lim s_n = s$. Some examples of convergent sequences that appear frequently are

$$(1.1) \quad \lim_{n \rightarrow \infty} \frac{1}{n^p} = 0, \quad 0 < p < \infty,$$

$$(1.2) \quad \lim_{n \rightarrow \infty} |z|^n = 0, \quad |z| < 1,$$

$$(1.3) \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1.$$

To prove (1.1) formally, we would for a given $\varepsilon > 0$ take N to be an integer satisfying $N > 1/\varepsilon^{1/p}$. Then for $n \geq N$ we have $n^p \geq N^p > 1/\varepsilon$, and $1/n^p < \varepsilon$. To prove (1.2) formally, we would take N to be an integer satisfying $N > (\log \varepsilon)/(\log |z|)$. To prove (1.3) formally, we resort to a trick. Let $t_n = \sqrt[n]{n} - 1$. We estimate t_n from the binomial expansion

$$n = (1 + t_n)^n = 1 + nt_n + \frac{n(n-1)}{2}t_n^2 + \cdots + t_n^n \geq \frac{n(n-1)}{2}t_n^2.$$

This yields $t_n^2 < 2/(n-1)$. Thus $|t_n| = |\sqrt[n]{n} - 1| < \varepsilon$ whenever $2/(n-1) < \varepsilon^2$, that is, for $n > 1 + 2/\varepsilon^2$. For the formal definition we can take N to be any integer satisfying $N > 1 + 2/\varepsilon^2$.

We give some definitions and state some theorems, without proofs, that we will be using.

A sequence of complex numbers $\{s_n\}$ is said to be **bounded** if there is some number $R > 0$ such that $|s_n| \leq R$ for all n . In other words, the sequence is bounded if it is contained in some disk.

Theorem. A convergent sequence is bounded. Further, if $\{s_n\}$ and $\{t_n\}$ are sequences of complex numbers such that $s_n \rightarrow s$ and $t_n \rightarrow t$, then

- (a) $s_n + t_n \rightarrow s + t$,
- (b) $s_n t_n \rightarrow st$,
- (c) $s_n/t_n \rightarrow s/t$, provided that $t \neq 0$.

Thus the limit of a sum is the sum of the limits, the limit of a product is the product of the limits, and the limit of a quotient is the quotient of the limits, provided that the denominator is not 0.

Example. We can use these rules to evaluate the limit of a rational expression of the form

$$\lim_{n \rightarrow \infty} \frac{3n^2 + 2n - 1}{5n^2 - 4n + 8} = \frac{3}{5}.$$

As a preliminary trick, we divide numerator and denominator by the leading power, and rewrite the expression as

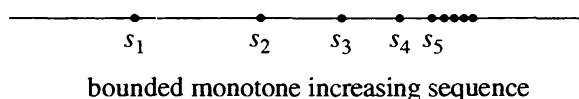
$$\frac{3 + (2/n) - 1/n^2}{5 - (4/n) + (8/n^2)}.$$

Since $1/n \rightarrow 0$ and $1/n^2 \rightarrow 0$, the sum and product statements show that the numerator converges to 3 and the denominator converges to 5. By the quotient statement, the quotient then converges to $\frac{3}{5}$.

The most useful criteria for convergence of sequences of real and complex numbers are gathered in the next several theorems. The first criterion is sometimes called the **in-between theorem**.

Theorem. If $r_n \leq s_n \leq t_n$, and if $r_n \rightarrow L$ and $t_n \rightarrow L$, then $s_n \rightarrow L$.

A sequence of real numbers $\{s_n\}$ is said to be **monotone increasing** if $s_{n+1} \geq s_n$ for all n , **monotone decreasing** if $s_{n+1} \leq s_n$ for all n , and **monotone** if it is either monotone increasing or decreasing. The following criterion is a version of the completeness axiom for the real numbers.



Theorem. A bounded monotone sequence of real numbers converges.

A sequence $\{s_n\}$ of real numbers can behave rather wildly. It is still possible to assign an “upper limit” to $\{s_n\}$, denoted by $\limsup s_n$, which is the largest possible limit of a subsequence of $\{s_n\}$. Our working definition is that $\limsup s_n$ is the unique extended real number S , $-\infty \leq S \leq +\infty$, such that if $t > S$, then $s_n \geq t$ for only finitely many indices n , while if $t < S$, then $s_n > t$ for infinitely many indices n . It is easy to see that any such S is unique. The existence of such an S can be deduced from the preceding theorem. In fact, the existence is equivalent to the completeness axiom of the real numbers.

A “lower limit” of the sequence $\{s_n\}$, denoted by $\liminf s_n$, is defined similarly. It satisfies

$$\liminf s_n = -\limsup(-s_n).$$

The sequence $\{s_n\}$ converges if and only if its $\limsup$ and $\liminf$ are finite and equal.

Example. The sequence $\{(-1)^n\}_{n=0}^{\infty} = \{+1, -1, +1, -1, \dots\}$ does not converge. Its upper and lower limits are

$$\limsup_{n \rightarrow \infty} (-1)^n = +1, \quad \liminf_{n \rightarrow \infty} (-1)^n = -1.$$

For complex sequences, the following simple criterion is used very often.

Theorem. *A sequence $\{s_k\}$ of complex numbers converges if and only if the corresponding sequences of real and imaginary parts of the s_k 's converge.*

We define a sequence of complex numbers $\{s_n\}$ to be a **Cauchy sequence** if the differences $s_n - s_m$ tend to 0 as n and m tend to ∞ . In the language of formal mathematics, this means that for any $\varepsilon > 0$, there exists $N \geq 1$ such that $|s_n - s_m| < \varepsilon$ if $m, n \geq N$. The following theorem is an equivalent form of the completeness axiom. It is important because it provides a means of determining whether a sequence is convergent without producing explicitly the limit of the sequence.

Theorem. *A sequence of complex numbers converges if and only if it is a Cauchy sequence.*

We say that a complex-valued function $f(z)$ **has limit L as z tends to z_0** if the values $f(z)$ are near L whenever z is near z_0 , $z \neq z_0$. The formal definition is that $f(z)$ has limit L as z tends to z_0 if for any $\varepsilon > 0$, there is $\delta > 0$ such that $|f(z) - L| < \varepsilon$ whenever z in the domain of $f(z)$ satisfies $0 < |z - z_0| < \delta$. In this case we write

$$\lim_{z \rightarrow z_0} f(z) = L,$$

or $f(z) \rightarrow L$ as $z \rightarrow z_0$. It is implicitly understood that there are points in the domain of $f(z)$ that are arbitrarily close to z_0 and different from z_0 . The definition can be rephrased in terms of convergent sequences.

Lemma. *The complex-valued function $f(z)$ has limit L as $z \rightarrow z_0$ if and only if $f(z_n) \rightarrow L$ for any sequence $\{z_n\}$ in the domain of $f(z)$ such that $z_n \neq z_0$ and $z_n \rightarrow z_0$.*

From the theorem on limits, we obtain easily the following.

Theorem. *If a function has a limit at z_0 , then the function is bounded near z_0 . Further, if $f(z) \rightarrow L$ and $g(z) \rightarrow M$ as $z \rightarrow z_0$, then as $z \rightarrow z_0$ we have*

- (a) $f(z) + g(z) \rightarrow L + M$,
- (b) $f(z)g(z) \rightarrow LM$,
- (c) $f(z)/g(z) \rightarrow L/M$, provided that $M \neq 0$.

We say that $f(z)$ is **continuous at z_0** if $f(z) \rightarrow f(z_0)$ as $z \rightarrow z_0$. A **continuous function** is a function that is continuous at each point of its domain. The preceding theorem shows that sums and products of continuous functions are continuous, and so are quotients, provided that the denominator is not zero. Further, the composition of continuous functions is continuous.

Example. Any constant function is continuous. The coordinate function $f(z) = z$ is continuous. Thus any polynomial function $p(z) = a_n z^n + \cdots + a_1 z + a_0$ is continuous. Any rational function $p(z)/q(z)$ is continuous wherever the denominator $q(z)$ is not zero.

A useful strategy for showing that $f(z)$ is continuous at z_0 is to obtain an estimate of the form $|f(z) - f(z_0)| \leq C|z - z_0|$ for z near z_0 . This guarantees that $|f(z) - f(z_0)| < \varepsilon$ whenever $|z - z_0| < \varepsilon/C$, so that we can take $\delta = \varepsilon/C$ in the formal definition of limit.

Example. The estimates

$$|\operatorname{Re}(z - z_0)| \leq |z - z_0|,$$

$$|\operatorname{Im}(z - z_0)| \leq |z - z_0|,$$

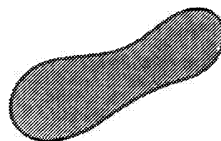
$$||z| - |z_0|| \leq |z - z_0|,$$

show respectively that the functions $\operatorname{Re}(z)$, $\operatorname{Im}(z)$, and $|z|$ are continuous.

A subset U of the complex plane is **open** if whenever $z \in U$, there is a disk centered at z that is contained in U .



open set: no boundary points



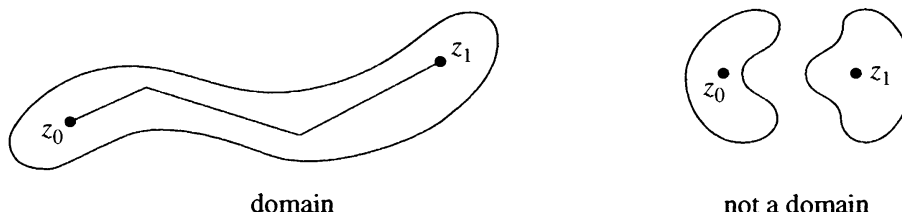
closed set: includes boundary

Any open disk $\{|z - z_0| < \rho\}$ is an open set. The closed disk $\{|z - z_0| \leq \rho\}$ is not an open set, since any open disk centered at a point on the boundary circle $\{|z - z_0| = \rho\}$ extends outside the closed disk.

In general, any set described by strict inequalities of continuous functions is open. For instance, the open upper half-plane is described by the strict inequality $\operatorname{Im}(z) > 0$, so that it is an open set. Other examples of open sets

described by strict inequalities are the open sector $\{\theta_0 < \arg z < \theta_1\}$, the open horizontal strip $\{-1 < \operatorname{Im} z < +1\}$, the open annulus $\{r < |z| < s\}$, and the punctured disk $\{0 < |z| < r\}$.

A subset D of the complex plane is a **domain** if D is open and if any two points of D can be connected by a broken line segment in D . Open half-planes, open disks, open sectors, open annuli, and open punctured disks are all domains. An example of an open set that is not a domain is the union of the open upper and lower half-planes, $U = \mathbb{C} \setminus \mathbb{R}$. It is impossible to connect a point in the upper half-plane to a point in the lower half-plane by a broken line segment that does not cross the real line.

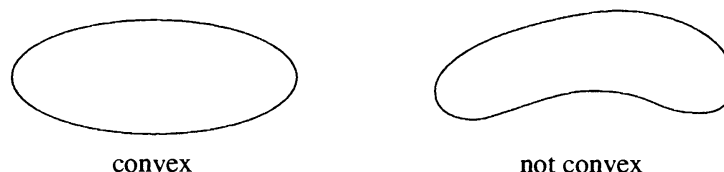


The most important property of domains for us is the following property, which actually characterizes domains (Exercise 18).

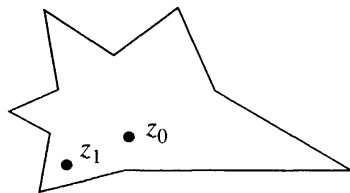
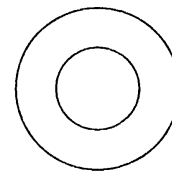
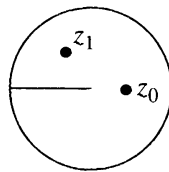
Theorem. *If $h(x, y)$ is a continuously differentiable function on a domain D such that $\nabla h = 0$ on D , then h is constant.*

This theorem is easy to justify. Since $\nabla h = 0$, the directional derivative of $h(x, y)$ in any direction is zero. Consequently, $h(x, y)$ is constant on any straight line segment contained in D , hence on any broken line segment. Since any two points of D can be joined by a broken line segment in D , $h(x, y)$ is constant on D .

A set is **convex** if whenever two points belong to the set, then the straight line segment joining the two points is contained in the set. An open or closed disk is convex, but a punctured disk is not convex.



A set is **star-shaped with respect to z_0** if whenever a point belongs to the set, then the straight line segment joining z_0 to the point is contained in the set. In other words, a set is star-shaped with respect to z_0 if every point of the set is visible from z_0 . Any convex set is star-shaped with respect to each of its points. The slit plane $\mathbb{C} \setminus (-\infty, 0]$ is star-shaped with respect to any point on the positive real axis. However, it is not convex, and it is not star-shaped with respect to any point not on the real axis.

star-shaped with respect to z_0 but not z_1 

not star-shaped

A **star-shaped domain** is a domain that is star-shaped with respect to one of its points. Thus $\mathbb{C} \setminus (-\infty, 0]$ is a star-shaped domain. Any convex domain is a star-shaped domain. An open annulus is not star-shaped.

A subset E of the complex plane is **closed** if it contains the limit of every convergent sequence in E . The closed disk $\{|z - z_0| \leq r\}$ is a closed set, since if $|s_n - z_0| \leq r$ and $s_n \rightarrow s$, then $|s - z_0| \leq r$.

Sets of the form $\{f(z) \geq c\}$ or $\{f(z) \leq c\}$, where $f(z)$ is a continuous real-valued function, are closed. Thus for instance the closed upper half-plane, consisting of points z such that $\operatorname{Re}(z) \geq 0$, is a closed set.

The **boundary** of a set E consists of points z such that every disk centered at z contains both points in E and points not in E . Thus a set is closed if it contains its boundary, and a set is open if it does not include any of its boundary points. For example, the boundary of the closed disk $\{|z - z_0| \leq r\}$ is its boundary circle $\{|z - z_0| = r\}$, and the boundary of the open disk $\{|z - z_0| < r\}$ is also the boundary circle.

A subset of the complex plane that is closed and bounded is said to be **compact**. A closed disk $\{|z - z_0| \leq r\}$ is compact, as is a closed interval $[a, b]$ on the real line. We will use the following important property of compact sets in our discussion of the maximum principle for harmonic and analytic functions.

Theorem. *A continuous real-valued function on a compact set attains its maximum.*

Exercises for II.1

1. Establish the following:

$$\begin{array}{ll} \text{(a)} \lim_{n \rightarrow \infty} \frac{n}{n+1} = 1 & \text{(c)} \lim_{n \rightarrow \infty} \frac{2n^p + 5n + 1}{n^p + 3n + 1} = 2, \quad p > 1 \\ \text{(b)} \lim_{n \rightarrow \infty} \frac{n}{n^2 + 1} = 0 & \text{(d)} \lim_{n \rightarrow \infty} \frac{z^n}{n!} = 0, \quad z \in \mathbb{C}. \end{array}$$

2. For which values of z is the sequence $\{z^n\}_{n=1}^{\infty}$ bounded? For which values of z does the sequence converge to 0?

3. Show that $\{n^n z^n\}$ converges only for $z = 0$.

4. Show that $\lim_{N \rightarrow \infty} \frac{N!}{N^k(N-k)!} = 1, \quad k \geq 0.$

5. Show that the sequence

$$b_n = 1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{n} - \log n, \quad n \geq 1,$$

is decreasing, while the sequence $a_n = b_n - 1/n$ is increasing. Show that the sequences both converge to the same limit γ . Show that $\frac{1}{2} < \gamma < \frac{3}{5}$. *Remark.* The limit of the sequence is called **Euler's constant**. It is not known whether Euler's constant is a rational number or an irrational number.

6. For a complex number α , we define the binomial coefficient “ α choose n ” by

$$\binom{\alpha}{0} = 1, \quad \binom{\alpha}{n} = \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}, \quad n \geq 1.$$

Show the following.

- (a) For each α , the sequence $\binom{\alpha}{n}$ is bounded.
 - (b) $\binom{\alpha}{n} \rightarrow 0$ if and only if $\operatorname{Re} \alpha > -1$.
 - (c) If $\alpha \neq 0, 1, 2, \dots$, then $\binom{\alpha}{n+1} / \binom{\alpha}{n} \rightarrow -1$.
 - (d) If $\operatorname{Re} \alpha \leq -1$, $\alpha \neq -1$, then $\left| \binom{\alpha}{n+1} \right| > \left| \binom{\alpha}{n} \right|$ for all $n \geq 0$.
 - (e) If $\operatorname{Re} \alpha > -1$ and α is not an integer, then $\left| \binom{\alpha}{n+1} \right| < \left| \binom{\alpha}{n} \right|$ for n large.
7. Define $x_0 = 0$, and define by induction $x_{n+1} = x_n^2 + \frac{1}{4}$ for $n \geq 0$. Show that $x_n \rightarrow \frac{1}{2}$. *Hint.* Show that the sequence is bounded and monotone, and that any limit satisfies $x = x^2 + \frac{1}{4}$.
8. Show that if $s_n \rightarrow s$, then $|s_n - s_{n-1}| \rightarrow 0$.
9. Plot each sequence and determine its $\liminf$ and $\limsup$.
- | | |
|--------------------------------------|---|
| (a) $s_n = 1 + \frac{1}{n} + (-1)^n$ | (c) $s_n = \sin(\pi n/4)$ |
| (b) $s_n = (-n)^n$ | (d) $s_n = x^n$ ($x \in \mathbb{R}$ fixed) |
10. At what points are the following functions continuous? Justify your answer. (a) z , (b) $z/|z|$, (c) $z^2/|z|$, (d) $z^2/|z|^3$.
11. At what points does the function $\operatorname{Arg} z$ have a limit? Where is $\operatorname{Arg} z$ continuous? Justify your answer.
12. Let $h(z)$ be the restriction of the function $\operatorname{Arg} z$ to the lower half-plane $\{\operatorname{Im} z < 0\}$. At what points does $h(z)$ have a limit? What is the limit?

13. For which complex values of α does the principal value of z^α have a limit as z tends to 0? Justify your answer.
14. Let $h(t)$ be a continuous complex-valued function on the unit interval $[0, 1]$, and consider

$$H(z) = \int_0^1 \frac{h(t)}{t-z} dt.$$

Where is $H(z)$ defined? Where is $H(z)$ continuous? Justify your answer. *Hint.* Use the fact that if $|f(t) - g(t)| < \varepsilon$ for $0 \leq t \leq 1$, then $\int_0^1 |f(t) - g(t)| dt < \varepsilon$.

15. Which of the following sets are open subsets of $\mathbb{C}$? Which are closed? Sketch the sets. (a) The punctured plane $\mathbb{C} \setminus \{0\}$, (b) the exterior of the open unit disk in the plane, $\{|z| \geq 1\}$, (c) the exterior of the closed unit disk in the plane, $\{|z| > 1\}$, (d) the plane with the open unit interval removed, $\mathbb{C} \setminus (0, 1)$, (e) the plane with the closed unit interval removed, $\mathbb{C} \setminus [0, 1]$, (f) the semidisk $\{|z| < 1, \operatorname{Im}(z) \geq 0\}$, (g) the complex plane $\mathbb{C}$.
16. Show that the slit plane $\mathbb{C} \setminus (-\infty, 0]$ is star-shaped but not convex. Show that the slit plane $\mathbb{C} \setminus [-1, 1]$ is not star-shaped. Show that a punctured disk is not star-shaped.
17. Show that a set is convex if and only if it is star-shaped with respect to each of its points.
18. Show that the following are equivalent for an open subset U of the complex plane.
 - (a) Any two points of U can be joined by a path consisting of straight line segments parallel to the coordinate axes.
 - (b) Any continuously differentiable function $h(x, y)$ on U such that $\nabla h = 0$ is constant.
 - (c) If V and W are disjoint open subsets of U such that $U = V \cup W$, then either $U = V$ or $U = W$. *Remark.* In the context of topological spaces, this latter property is taken as the definition of connectedness.
19. Give a proof of the fundamental theorem of algebra along the following lines. Show that if $p(z)$ is a nonconstant polynomial, then $|p(z)|$ attains its minimum at some point $z_0 \in \mathbb{C}$. Assume that the minimum is attained at $z_0 = 0$, and that $p(z) = 1 + az^m + \cdots$, where $m \geq 1$ and $a \neq 0$. Contradict the minimality by showing that $|P(\varepsilon e^{i\theta_0})| < 1$ for an appropriate choice of θ_0 .

2. Analytic Functions

If the development in this section has a familiar ring, it should. The basic definitions and rules for the complex derivative are exactly the same as those for the usual derivative in elementary calculus. The only difference is that multiplication and division are now performed with complex numbers instead of real numbers.

A complex-valued function $f(z)$ is **differentiable** at z_0 if the difference quotients

$$(2.1) \quad \frac{f(z) - f(z_0)}{z - z_0}$$

have a limit as $z \rightarrow z_0$. The limit is denoted by $f'(z_0)$, or by $\frac{df}{dz}(z_0)$, and we refer to it as the **complex derivative** of $f(z)$ at z_0 . Thus

$$(2.2) \quad \frac{df}{dz}(z_0) = f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}.$$

Example. A constant function $f(z) = c$ has derivative $f'(z_0) = 0$ at any point z_0 . In this case the difference quotients (2.1) are all zero, so that the limit is also 0.

It is often useful to write the difference quotient (2.1) in the form

$$(2.3) \quad \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z},$$

so that $z - z_0$ is replaced by Δz . The formula for the complex derivative becomes

$$(2.4) \quad f'(z_0) = \lim_{\Delta z \rightarrow 0} \frac{f(z_0 + \Delta z) - f(z_0)}{\Delta z}.$$

Occasionally we use z instead of z_0 in the expression (2.4).

Example. The power function $f(z) = z^m$ has derivative $f'(z) = mz^{m-1}$. In this case the binomial expansion

$$(z + \Delta z)^m = z^m + mz^{m-1}\Delta z + \frac{m(m-1)}{2}z^{m-2}(\Delta z)^2 + \cdots + (\Delta z)^m$$

yields

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = mz^{m-1} + \frac{m(m-1)}{2}z^{m-2}\Delta z + \cdots + (\Delta z)^{m-1},$$

which has limit mz^{m-1} as $\Delta z \rightarrow 0$.

Example. The function $f(z) = \bar{z}$ is not differentiable at any point z . In this case the difference quotient (2.3) becomes

$$((z + \Delta z) - \bar{z}) / \Delta z = \overline{\Delta z} / \Delta z.$$

If $\Delta z = \varepsilon$ is real, then this difference quotient is equal to 1, whereas if $\Delta z = i\varepsilon$ is imaginary, then the difference quotient is equal to -1 . Thus the difference quotients do not have a limit as $\Delta z \rightarrow 0$.

The various properties of the complex derivative can be developed in exactly the same way as the properties of the usual derivative.

Theorem. *If $f(z)$ is differentiable at z_0 , then $f(z)$ is continuous at z_0 .*

This follows from the sum and product rules for limits. We write

$$f(z) = f(z_0) + \left(\frac{f(z) - f(z_0)}{z - z_0} \right) (z - z_0).$$

Since the difference quotient tends to $f'(z_0)$ as $z \rightarrow z_0$, and $z - z_0$ tends to 0 as $z \rightarrow z_0$, the product on the right tends to 0, and consequently, $f(z) \rightarrow f(z_0)$ as $z \rightarrow z_0$.

The complex derivative satisfies the usual rules for differentiating sums, products, and quotients. The rules are

$$(2.5) \quad (cf)'(z) = cf'(z),$$

$$(2.6) \quad (f + g)'(z) = f'(z) + g'(z),$$

$$(2.7) \quad (fg)'(z) = f(z)g'(z) + f'(z)g(z),$$

$$(2.8) \quad (f/g)'(z) = \frac{g(z)f'(z) - f(z)g'(z)}{g(z)^2}, \quad g(z) \neq 0.$$

Here we are assuming that $f(z)$ and $g(z)$ are differentiable at z , and that c is any complex constant. The conclusion is that $cf(z)$, $f(z) + g(z)$, $f(z)g(z)$, and, provided that $g(z) \neq 0$, also $f(z)/g(z)$ are all differentiable at z and satisfy the rules (2.5) to (2.8) listed above. The proofs depend on the theorems for limits of sums, products, and quotients. For instance, to establish the product rule (2.7) we begin with the usual trick and rewrite the difference quotient $[(fg)(z + \Delta z) - (fg)(z)]/\Delta z$ for the product as

$$f(z + \Delta z) \frac{g(z + \Delta z) - g(z)}{\Delta z} + \frac{f(z + \Delta z) - f(z)}{\Delta z} g(z).$$

We now take a limit as $\Delta z \rightarrow 0$ and apply the rules for limits of sums and products, and we obtain (2.7).

The identities (2.5) and (2.6) express the fact that complex differentiation is a linear operation. Note that (2.5) is a consequence of the product rule (2.7) and the fact that the derivative of a constant function c is 0.

To establish the identity (2.8), it suffices to establish the simpler identity

$$(2.9) \quad (1/g)'(z) = -g'(z)/g(z)^2, \quad g(z) \neq 0,$$

and then to apply the product rule.

Example. Any polynomial

$$p(z) = a_n z^n + a_{n-1} z^{n-1} + \cdots + a_1 z + a_0$$

has a complex derivative, which is given by the usual formula

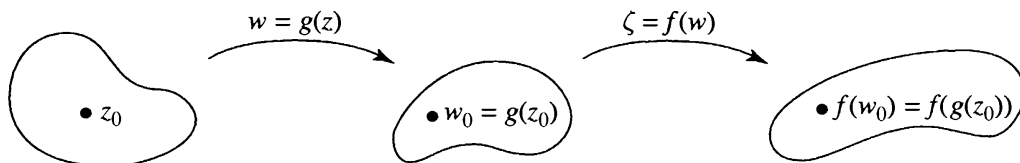
$$p'(z) = n a_n z^{n-1} + (n-1) a_{n-1} z^{n-2} + \cdots + a_1.$$

This follows from the linearity rules (2.5) and (2.6), since z^m is differentiable with derivative mz^{m-1} . Further, any rational function $p(z)/q(z)$ is differentiable at all points z except for the (finitely many) zeros of $q(z)$.

The chain rule is also valid for the complex derivative. We give a careful statement and proof.

Theorem (Chain Rule). Suppose that $g(z)$ is differentiable at z_0 , and suppose that $f(w)$ is differentiable at $w_0 = g(z_0)$. Then the composition $(f \circ g)(z) = f(g(z))$ is differentiable at z_0 and

$$(2.10) \quad (f \circ g)'(z_0) = f'(g(z_0))g'(z_0).$$



A useful mnemonic device for remembering the chain rule is

$$\frac{df}{dz} = \frac{df}{dw} \frac{dw}{dz},$$

where we have written $w = g(z)$. *Danger!* We regard f on the right-hand side as a function of w , and we regard f on the left-hand side as the function $f(g(z))$ of z . The mnemonic device can be justified by the proof, which involves multiplying and dividing by Δw . The proof goes as follows.

We consider two cases. For the first case, we assume that $g'(z_0) \neq 0$. Then $g(z) \neq g(z_0)$ for $0 < |z - z_0| < \varepsilon$, so we are justified in writing

$$(2.11) \quad \frac{f(g(z)) - f(g(z_0))}{z - z_0} = \frac{f(g(z)) - f(g(z_0))}{g(z) - g(z_0)} \frac{g(z) - g(z_0)}{z - z_0}.$$

Since $g(z)$ is differentiable at z_0 , it is continuous at z_0 , that is, $g(z) \rightarrow g(z_0)$ as $z \rightarrow z_0$. Consequently,

$$\frac{f(g(z)) - f(g(z_0))}{g(z) - g(z_0)} \rightarrow f'(g(z_0))$$

as $z \rightarrow z_0$. Thus we can pass to the limit in (2.11), and we obtain (2.10).

For the second case, we assume that $g'(z_0) = 0$. Since $f(w)$ is differentiable at w_0 , the difference quotients $(f(w) - f(w_0))/(w - w_0)$ are bounded near w_0 , say

$$\left| \frac{f(w) - f(w_0)}{w - w_0} \right| \leq C$$

for some constant C and $0 < |w - w_0| < \varepsilon$. Hence $|f(g(z)) - f(g(z_0))| \leq C|g(z) - g(z_0)|$ for z near z_0 , and consequently,

$$\left| \frac{f(g(z)) - f(g(z_0))}{z - z_0} \right| \leq C \left| \frac{g(z) - g(z_0)}{z - z_0} \right|.$$

Since the right-hand side tends to 0 as $z \rightarrow z_0$, we obtain $(f \circ g)'(z_0) = 0$. Thus both sides of (2.10) are 0, and in particular, the identity (2.10) holds.

Example. Suppose $f(w) = 1/w$, and $g(z) = z^2 - 1$. Then $f(g(z)) = 1/(z^2 - 1)$. Using $f'(w) = -1/w^2$ and $g'(z) = 2z$, we obtain from the chain rule

$$\frac{d}{dz} \frac{1}{z^2 - 1} = \left(-\frac{1}{w^2} \Big|_{w=z^2-1} \right) (2z) = -\frac{2z}{(z^2 - 1)^2}, \quad z \neq \pm 1.$$

This is, of course, the same as the result we would have obtained by applying the quotient rule. More generally, the rule (2.9) follows from the chain rule and the formula for the derivative for $1/w$.

Now we turn to the definition of the class of functions that is the main subject of complex analysis. As usual, all our functions will be complex-valued functions defined on a subset of the complex plane.

Definition. A function $f(z)$ is **analytic on the open set** U if $f(z)$ is (complex) differentiable at each point of U and the complex derivative $f'(z)$ is continuous on U .

We have seen that any polynomial in z has a complex derivative at any point, and the complex derivative is a polynomial, hence continuous. Thus any polynomial in z is analytic on the entire complex plane. Rational functions are analytic wherever they are finite.

More generally, the rules established for complex derivatives show that sums and products of analytic functions are analytic. Quotients of analytic functions are analytic wherever the denominator does not vanish.

An example of a function that is not analytic is $f(z) = \bar{z}$, which does not have a complex derivative at any point.

The requirement that $f'(z)$ be continuous is a nuisance to verify. The student will be happy to learn that this condition is redundant. In Chapter IV we will prove Goursat's theorem, that if $f'(z)$ exists at each point of an open set U , then $f'(z)$ is automatically continuous on U . Meanwhile, the student who is willing to take this theorem on faith need not check the

continuity of $f'(z)$, though in all cases we will treat, where $f'(z)$ can be shown to exist, it will also be apparent that $f'(z)$ is continuous.

Exercises for II.2

1. Find the derivatives of the following functions.
 (a) $z^2 - 1$ (c) $(z^2 - 1)^n$ (e) $1/(z^2 + 3)$ (g) $(az + b)/(cz + d)$
 (b) $z^n - 1$ (d) $1/(1 - z)$ (f) $z/(z^3 - 5)$ (h) $1/(cz + d)^2$
2. Show that

$$1 + 2z + 3z^2 + \cdots + nz^{n-1} = \frac{1 - z^n}{(1 - z)^2} - \frac{nz^n}{1 - z}.$$

3. Show from the definition that the functions $x = \operatorname{Re} z$ and $y = \operatorname{Im} z$ are not complex differentiable at any point.
4. Suppose $f(z) = az^2 + bz\bar{z} + c\bar{z}^2$, where a , b , and c are fixed complex numbers. By differentiating $f(z)$ by hand, show that $f(z)$ is complex differentiable at z if and only if $bz + 2c\bar{z} = 0$. Where is $f(z)$ analytic?
5. Show that if f is analytic on D , then $g(z) = \overline{f(\bar{z})}$ is analytic on the reflected domain $D^* = \{\bar{z} : z \in D\}$, and $g'(z) = \overline{f'(\bar{z})}$.
6. Let $h(t)$ be a continuous complex-valued function on the unit interval $[0, 1]$, and define

$$H(z) = \int_0^1 \frac{h(t)}{t - z} dt, \quad z \in \mathbb{C} \setminus [0, 1].$$

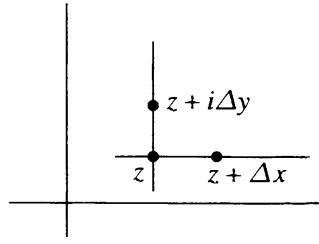
Show that $H(z)$ is analytic and compute its derivative. *Hint.* Differentiate by hand; that is, use the defining identity (2.4) of the complex derivative.

3. The Cauchy-Riemann Equations

Suppose $f = u + iv$ is analytic on a domain D . Fix a point $z \in D$. We will compute the complex derivative

$$f'(z) = \lim_{\Delta z \rightarrow 0} \frac{f(z + \Delta z) - f(z)}{\Delta z}$$

in two different ways, first by letting $z + \Delta z$ tend to z along the horizontal line through z (that is, $\Delta z = \Delta x$ real), then by letting $z + \Delta z$ tend to z along the vertical line through z (that is, $\Delta z = i\Delta y$ imaginary). This yields two expressions for $f'(z)$, which lead to the Cauchy-Riemann equations.



Expressing the difference quotient in terms of u and v and setting $\Delta z = \Delta x$, we obtain

$$\begin{aligned} \frac{f(z + \Delta x) - f(z)}{\Delta x} &= \frac{u(x + \Delta x, y) + iv(x + \Delta x, y) - (u(x, y) + iv(x, y))}{\Delta x} \\ &= \frac{u(x + \Delta x, y) - u(x, y)}{\Delta x} + i \frac{v(x + \Delta x, y) - v(x, y)}{\Delta x}. \end{aligned}$$

Passing to the limit, we see that the x -derivatives of u and v exist, and

$$(3.1) \quad f'(z) = \frac{\partial u}{\partial x}(x, y) + i \frac{\partial v}{\partial x}(x, y), \quad z = x + iy.$$

Since $f'(z)$ is continuous, (3.1) shows that the x -derivatives of u and v are continuous.

Next we set $\Delta z = i\Delta y$, and we play the same game. The difference quotient becomes

$$\begin{aligned} \frac{f(z + i\Delta y) - f(z)}{i\Delta y} &= \frac{u(x, y + \Delta y) + iv(x, y + \Delta y) - (u(x, y) + iv(x, y))}{i\Delta y} \\ &= \frac{v(x, y + \Delta y) - v(x, y)}{\Delta y} - i \frac{u(x, y + \Delta y) - u(x, y)}{\Delta y}. \end{aligned}$$

Passing to the limit as before, we see that the y -derivatives of u and v are continuous and satisfy

$$(3.2) \quad f'(z) = \frac{\partial v}{\partial y}(x, y) - i \frac{\partial u}{\partial y}(x, y), \quad z = x + iy.$$

Now we have two expressions, (3.1) and (3.2), for $f'(z)$. We equate their real and imaginary parts, and we obtain

$$(3.3) \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

These equations are called the **Cauchy-Riemann equations** for u and v . We have proved half of the following theorem.

Theorem. *Let $f = u + iv$ be defined on a domain D in the complex plane, where u and v are real-valued. Then $f(z)$ is analytic on D if and only if $u(x, y)$ and $v(x, y)$ have continuous first-order partial derivatives that satisfy the Cauchy-Riemann equations (3.3).*

It remains to be shown that if the partial derivatives of u and v exist, are continuous, and satisfy the Cauchy-Riemann equations, then $f = u + iv$ is analytic. For this, we use Taylor's theorem. Fix $z \in D$. Taylor's theorem with remainder provides an approximation

$$u(x + \Delta x, y + \Delta y) = u(x, y) + \frac{\partial u}{\partial x}(x, y)\Delta x + \frac{\partial u}{\partial y}(x, y)\Delta y + R(\Delta x, \Delta y),$$

where $R(\Delta x, \Delta y)/|\Delta z|$ has limit 0 as Δz approaches 0. (The continuity of the first-order partial derivatives of u must be used to obtain the estimate for R .) Similarly,

$$v(x + \Delta x, y + \Delta y) = v(x, y) + \frac{\partial v}{\partial x}(x, y)\Delta x + \frac{\partial v}{\partial y}(x, y)\Delta y + S(\Delta x, \Delta y),$$

where $S(\Delta x, \Delta y)/|\Delta z| \rightarrow 0$ as $\Delta z \rightarrow 0$. Thus

$$\begin{aligned} f(z + \Delta z) &= f(z) + \frac{\partial u}{\partial x}(x, y)\Delta x + \frac{\partial u}{\partial y}(x, y)\Delta y + R(\Delta z) \\ &\quad + i\frac{\partial v}{\partial x}(x, y)\Delta x + i\frac{\partial v}{\partial y}(x, y)\Delta y + iS(\Delta z). \end{aligned}$$

If we use the Cauchy-Riemann equations to replace the y -derivatives by x -derivatives, and we use $\Delta x + i\Delta y = \Delta z$, a minor miracle occurs. The identity becomes

$$f(z + \Delta z) = f(z) + \left(\frac{\partial u}{\partial x}(x, y) + i\frac{\partial v}{\partial x}(x, y) \right) \Delta z + R(\Delta z) + iS(\Delta z).$$

Thus

$$\frac{f(z + \Delta z) - f(z)}{\Delta z} = \frac{\partial u}{\partial x}(x, y) + i\frac{\partial v}{\partial x}(x, y) + \frac{R(\Delta z) + iS(\Delta z)}{\Delta z},$$

which tends to

$$\frac{\partial u}{\partial x}(x, y) + i\frac{\partial v}{\partial x}(x, y)$$

as Δz tends to 0. This shows that $f'(z)$ exists and is given by (3.1), so that $f'(z)$ is continuous, and thus $f(z)$ is analytic. Both directions of the theorem are now established.

Example. The functions $u(x, y) = x$ and $v(x, y) = y$, corresponding to $z = x + iy$, satisfy the Cauchy-Riemann equations, since

$$\frac{\partial u}{\partial x} = 1 = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = 0 = -\frac{\partial v}{\partial x}.$$

The functions $u(x, y) = x$ and $v(x, y) = -y$, corresponding to $\bar{z} = x - iy$, do not satisfy the Cauchy-Riemann equations, since

$$\frac{\partial u}{\partial x} = 1, \quad \frac{\partial v}{\partial y} = -1.$$

We may use the Cauchy-Riemann equations to show that the function e^z is analytic and satisfies

$$\frac{d}{dz}e^z = e^z.$$

In this case, $u(x, y) = e^x \cos y$ and $v(x, y) = e^x \sin y$. We check that the Cauchy-Riemann equations hold:

$$\begin{aligned}\frac{\partial u}{\partial x} &= \frac{\partial}{\partial x} e^x \cos y = e^x \cos y = \frac{\partial}{\partial y} e^x \sin y = \frac{\partial v}{\partial y}, \\ \frac{\partial u}{\partial y} &= \frac{\partial}{\partial y} e^x \cos y = -e^x \sin y = -\frac{\partial}{\partial x} e^x \sin y = -\frac{\partial v}{\partial x}.\end{aligned}$$

Thus $f(z) = e^z$ is analytic, and (3.1) yields

$$f'(z) = \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} = e^x \cos y + i e^x \sin y = e^z.$$

From the chain rule we deduce further that any complex exponential function of the form e^{az} , where a is a complex constant, is analytic and satisfies

$$\frac{d}{dz}e^{az} = ae^z.$$

Linear combinations, such as $\sin z$ and $\cos z$, of complex exponential functions are also analytic, and the usual formulae for the derivatives hold:

$$\begin{aligned}\frac{d}{dz} \sin z &= \cos z, \\ \frac{d}{dz} \cos z &= -\sin z, \\ \frac{d}{dz} \sinh z &= \cosh z, \\ \frac{d}{dz} \cosh z &= \sinh z.\end{aligned}$$

To verify the formula for the derivative of $\sin z$, for instance, we compute

$$\frac{d}{dz} \sin z = \frac{d}{dz} \frac{e^{iz} - e^{-iz}}{2i} = \frac{ie^{iz} + ie^{-iz}}{2i} = \cos z.$$

Two important consequences of the Cauchy-Riemann equations and the equations for $f'(z)$ are as follows.

Theorem. *If $f(z)$ is analytic on a domain D , and if $f'(z) = 0$ on D , then $f(z)$ is constant.*

In this case, the equations (3.1) and (3.2) yield

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} = \frac{\partial v}{\partial x} = \frac{\partial v}{\partial y} = 0.$$

Since D is a domain, the theorem in Section 1 shows that u and v are constant on D . Thus $f = u + iv$ is constant.

Theorem. *If $f(z)$ is analytic and real-valued on a domain D , then $f(z)$ is constant.*

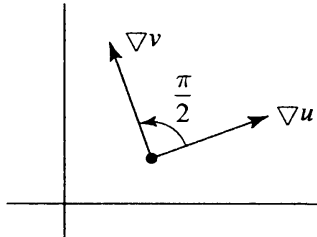
In this case, $v = 0$ on D , and the Cauchy-Riemann equations become

$$\frac{\partial u}{\partial x} = 0, \quad \frac{\partial u}{\partial y} = 0.$$

Since D is a domain, u is constant in D .

Exercises for II.3

- Find the derivatives of the following functions.
 (a) $\tan z = \frac{\sin z}{\cos z}$ (b) $\tanh z = \frac{\sinh z}{\cosh z}$ (c) $\sec z = 1/\cos z$
- Show that $u = \sin x \sinh y$ and $v = \cos x \cosh y$ satisfy the Cauchy-Riemann equations. Do you recognize the analytic function $f = u + iv$? (Determine its complex form.)
- Show that if f and $\bar{f}$ are both analytic on a domain D , then f is constant.
- Show that if f is analytic on a domain D , and if $|f|$ is constant, then f is constant. *Hint.* Write $\bar{f} = |f|^2/f$.
- If $f = u + iv$ is analytic, then $|\nabla u| = |\nabla v| = |f'|$.
- If $f = u + iv$ is analytic on D , then ∇v is obtained by rotating ∇u by 90° . In particular, ∇u and ∇v are orthogonal.



- Sketch the vector fields ∇u and ∇v for the following functions $f = u + iv$. (a) iz , (b) z^2 , (c) $1/z$.
- Derive the polar form of the Cauchy-Riemann equations for u and v :

$$\frac{\partial u}{\partial r} = \frac{1}{r} \frac{\partial v}{\partial \theta}, \quad \frac{\partial u}{\partial \theta} = -r \frac{\partial v}{\partial r}.$$

Check that for any integer m , the functions $u(re^{i\theta}) = r^m \cos(m\theta)$ and $v(re^{i\theta}) = r^m \sin(m\theta)$ satisfy the Cauchy-Riemann equations.

4. Inverse Mappings and the Jacobian

Let $f = u + iv$ be analytic on a domain D . We may regard D as a domain in the Euclidean plane $\mathbb{R}^2$ and f as a map from D to $\mathbb{R}^2$ with components $(u(x, y), v(x, y))$. The Jacobian matrix of this map is

$$J_f = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{pmatrix},$$

and the determinant of the Jacobian matrix is

$$\det J_f = \frac{\partial u}{\partial x} \frac{\partial v}{\partial y} - \frac{\partial u}{\partial y} \frac{\partial v}{\partial x}.$$

If we use the Cauchy-Riemann equations to replace the y -derivatives by x -derivatives, we obtain

$$\det J_f = \left(\frac{\partial u}{\partial x} \right)^2 + \left(\frac{\partial v}{\partial x} \right)^2 = \left| \frac{\partial u}{\partial x} + i \frac{\partial v}{\partial x} \right|^2.$$

By equation (3.2), this is equal to $|f'(z)|^2$. We have shown the following.

Theorem. *If $f(z)$ is analytic, then its Jacobian matrix J_f (as a map from $\mathbb{R}^2$ to $\mathbb{R}^2$) has determinant*

$$\det J_f(z) = |f'(z)|^2.$$

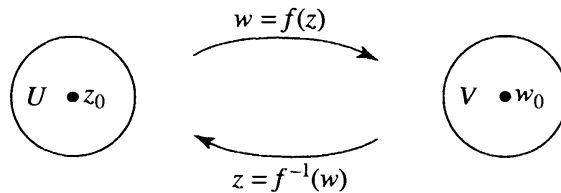
Now we can invoke the inverse function theorem from multivariable calculus, and this leads to the following.

Theorem. *Suppose $f(z)$ is analytic on a domain D , $z_0 \in D$, and $f'(z_0) \neq 0$. Then there is a (small) disk $U \subset D$ containing z_0 such that $f(z)$ is one-to-one on U , the image $V = f(U)$ of U is open, and the inverse function*

$$f^{-1} : V \longrightarrow U$$

is analytic and satisfies

$$(4.1) \quad (f^{-1})'(f(z)) = 1/f'(z), \quad z \in U.$$



All of the assertions of this theorem are consequences of the inverse function theorem, except for the assertions concerning the analyticity of f^{-1} . To check that f^{-1} is analytic, we write $g = f^{-1}$ on U and differentiate by hand. Fix $w, w_1 \in U$ with $w \neq w_1$, and set $z = g(w)$, $z_1 = g(w_1)$. Then $z \neq z_1$, $f(z) = w$, $f(z_1) = w_1$, and we have

$$\frac{g(w) - g(w_1)}{w - w_1} = \frac{z - z_1}{f(z) - f(z_1)} = 1 / \left(\frac{f(z) - f(z_1)}{z - z_1} \right).$$

As w tends to w_1 , z tends to z_1 , and the right-hand side tends to $1/f'(z_1)$. Thus g is differentiable at w_1 , and $g'(w_1) = 1/f'(z_1)$, which is the required identity (4.1) at z_1 . Since $1/f'(z)$ is continuous, $(f^{-1})'$ is continuous, and thus f^{-1} is analytic.

If we write $w = g(z)$, the identity (4.1) becomes

$$\frac{dz}{dw} = \frac{1}{\frac{dw}{dz}},$$

which is the usual mnemonic device for remembering the derivative of the inverse function. The device is justified by the proof. *Danger!* Take care to evaluate the derivatives at the right points.

Once we know that f^{-1} is analytic, we can easily derive the formula (4.1) for the derivative from the chain rule. Since $f^{-1}(f(z)) = z$, the chain rule yields $(f^{-1})'(f(z))f'(z) = 1$, which is (4.1).

Example. The principal logarithm function $w = \text{Log } z$ is a continuous inverse for $z = e^w$ for $-\pi < \arg w < \pi$. Since e^w is analytic and $(e^w)' \neq 0$, the preceding theorem applies, with z and w interchanged. From that theorem we conclude that $\text{Log } z$ is analytic. If we use the chain rule to differentiate

$$z = e^{\text{Log } z},$$

we obtain

$$1 = e^{\text{Log } z} \frac{d}{dz}(\text{Log } z) = z \frac{d}{dz}(\text{Log } z).$$

Thus

$$(4.2) \quad \frac{d}{dz} \text{Log } z = \frac{1}{z}.$$

Any other continuous branch of the logarithm differs from the principal branch by a constant, hence has the same derivative.

Example. Any continuous branch of $\sqrt{z}$ is analytic, and

$$(4.3) \quad \frac{d}{dz} \sqrt{z} = \frac{1}{2\sqrt{z}},$$

where we use the same branch of $\sqrt{z}$ on both sides of the identity. To see this, note first that no continuous branch of $\sqrt{z}$ can be defined on a domain containing 0, so that $z \neq 0$ in (4.3). Each branch $w = \sqrt{z}$ satisfies $w^2 = z$. Since $(w^2)' = 2w$ is not zero for $w \neq 0$, the continuous inverse branch $\sqrt{z}$ is analytic. Differentiating $w^2 = z$, we obtain

$$2w \frac{dw}{dz} = 1, \quad \frac{dw}{dz} = \frac{1}{2w},$$

which is (4.3).

We will give in Section VIII.4 another proof of the existence and analyticity of the inverse of an analytic function, which does not depend on the inverse function theorem from calculus but rather on residue theory. That proof will provide an explicit integral representation formula for the inverse function.

Exercises for II.4

1. Sketch the gradient vector fields ∇u and ∇v for (a) $u + iv = e^z$, (b) $u + iv = \operatorname{Log} z$.
2. Let a be a complex number, $a \neq 0$, and let $f(z)$ be an analytic branch of z^a on $\mathbb{C} \setminus (-\infty, 0]$. Show that $f'(z) = af(z)/z$. (Thus $f'(z) = az^{a-1}$, where we pick the branch of z^{a-1} that corresponds to the original branch of z^a divided by z .)
3. Consider the branch of $f(z) = \sqrt{z(1-z)}$ on $\mathbb{C} \setminus [0, 1]$ that has positive imaginary part at $z = 2$. What is $f'(z)$? Be sure to specify the branch of the expression for $f'(z)$.
4. Recall that the principal branch of the inverse tangent function was defined on the complex plane with two slits on the imaginary axis by

$$\operatorname{Tan}^{-1} z = \frac{1}{2i} \operatorname{Log} \left(\frac{1+iz}{1-iz} \right), \quad z \notin (-i\infty, -i] \cup [i, i\infty).$$

Find the derivative of $\operatorname{Tan}^{-1} z$. Find the derivative of $\tan^{-1} z$ for any analytic branch of the function defined on a domain D .

5. Recall that $\cos^{-1}(z) = -i \log[z \pm \sqrt{z^2 - 1}]$. Suppose $g(z)$ is an analytic branch of $\cos^{-1}(z)$, defined on a domain D . Find $g'(z)$. Do different branches of $\cos^{-1}(z)$ have the same derivative?
6. Suppose $h(z)$ is an analytic branch of $\sin^{-1}(z)$, defined on a domain D . Find $h'(z)$. Do different branches of $\sin^{-1}(z)$ have the same derivative?
7. Let $f(z)$ be a bounded analytic function, defined on a bounded domain D in the complex plane, and suppose that $f(z)$ is one-to-

one. Show that the area of $f(D)$ is given by

$$\text{Area}(f(D)) = \iint_D |f'(z)|^2 dx dy.$$

8. Sketch the image of the circle $\{|z - 1| \leq 1\}$ under the map $w = z^2$. Compute the area of the image.
9. Compute

$$\iint_D |f'(z)|^2 dx dy,$$

for $f(z) = z^2$ and D the open unit disk $\{|z| \leq 1\}$. Interpret your answer in terms of areas.

10. For smooth functions g and h defined on a bounded domain U , we define the **Dirichlet form** $D_U(g, h)$ by

$$D_U(g, h) = \iint_U \left[\frac{\partial g}{\partial x} \frac{\partial \bar{h}}{\partial x} + \frac{\partial g}{\partial y} \frac{\partial \bar{h}}{\partial y} \right] dx dy.$$

Show that if $z = f(\zeta)$ is a one-to-one analytic function from the bounded domain V onto U , then

$$D_U(g, h) = D_V(g \circ f, h \circ f).$$

Remark. This shows that the Dirichlet form is a “conformal invariant.”

5. Harmonic Functions

The equation

$$\frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_n^2} = 0$$

is called **Laplace’s equation**. The operator

$$\Delta = \frac{\partial^2}{\partial x_1^2} + \cdots + \frac{\partial^2}{\partial x_n^2}$$

is called the **Laplacian**. In terms of this operator, Laplace’s equation becomes simply $\Delta u = 0$. Smooth functions $u(x_1, \dots, x_n)$ that satisfy Laplace’s equation are called **harmonic functions**. Laplace’s equation is one of the most important partial differential equations of mathematical physics. Some indication of the applications will be given in Chapter III.

We will be concerned with harmonic functions of two variables, that is, solutions of

$$\Delta u = \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0.$$

We say that a function $u(x, y)$ is **harmonic** if all its first- and second-order partial derivatives exist and are continuous and satisfy Laplace's equation. In the case of functions of two variables, there is an intimate connection between analytic functions and harmonic functions.

Theorem. *If $f = u + iv$ is analytic, and the functions u and v have continuous second-order partial derivatives, then u and v are harmonic.*

The second hypothesis of the theorem is redundant. We will show in Chapter IV that an analytic function has continuous partial derivatives of all orders.

The harmonicity of u and v is a simple consequence of the Cauchy-Riemann equations,

$$(5.1) \quad \frac{\partial u}{\partial x} = \frac{\partial v}{\partial y},$$

$$(5.2) \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}.$$

Using these, we obtain

$$\frac{\partial^2 u}{\partial x^2} = \frac{\partial}{\partial x} \frac{\partial v}{\partial y} = \frac{\partial}{\partial y} \frac{\partial v}{\partial x} = -\frac{\partial^2 u}{\partial y^2},$$

which shows that u is harmonic. The verification that v is harmonic is the same.

If u is harmonic on a domain D , and v is a harmonic function such that $u + iv$ is analytic, we say that v is a **harmonic conjugate** of u . The harmonic conjugate v is unique, up to adding a constant. Indeed, if v_0 is another harmonic conjugate for u , so that $u + iv_0$ is also analytic, then the difference $i(v - v_0)$ is analytic, and $v - v_0$ is a real-valued analytic function, hence constant on D .

Exercise. Show that $u(x, y) = xy$ is harmonic, and find a harmonic conjugate for u .

Solution. We have

$$\frac{\partial^2}{\partial x^2} xy = 0 = -\frac{\partial^2}{\partial y^2} xy,$$

so that xy is harmonic. To find a harmonic conjugate v , we solve the Cauchy-Riemann equations. From (5.1) we have

$$\frac{\partial u}{\partial x} = y = \frac{\partial v}{\partial y}.$$

Thus

$$v(x, y) = \frac{y^2}{2} + h(x),$$

where $h(x)$ depends only on x and not on y . Equation (5.2) becomes $x = -h'(x)$, which has solution $h(x) = -x^2/2 + C$. Thus

$$v(x, y) = \frac{y^2}{2} - \frac{x^2}{2} + C,$$

where C is a constant. The analytic function $f = u + iv$ is given by

$$f(z) = -i\frac{z^2}{2} + iC.$$

The method used above actually shows that any harmonic function on a rectangle with sides parallel to the axes has a harmonic conjugate on the rectangle. Indeed, let $u(x, y)$ be harmonic on such a rectangle D , and let (x_0, y_0) be any fixed point of D . If we integrate the first Cauchy-Riemann equation (5.1) along a vertical segment from y_0 to y , with x fixed, we obtain

$$v(x, y) = \int_{y_0}^y \frac{\partial u}{\partial x}(x, t) dt + h(x),$$

where $h(x)$ is the constant of integration with respect to y . Though $h(x)$ does not depend on y , it may depend on x . The second Cauchy-Riemann equation (5.2) then becomes

$$\frac{\partial u}{\partial y}(x, y) = -\frac{\partial}{\partial x} \int_{y_0}^y \frac{\partial u}{\partial x}(x, t) dt - h'(x).$$

If we differentiate under the integral sign (as we may) and use Laplace's equation, we obtain

$$\begin{aligned} \frac{\partial u}{\partial y}(x, y) &= - \int_{y_0}^y \frac{\partial^2 u}{\partial x^2}(x, t) dt - h'(x) = \int_{y_0}^y \frac{\partial^2 u}{\partial y^2}(x, t) dt - h'(x) \\ &= \frac{\partial u}{\partial y}(x, y) - \frac{\partial u}{\partial y}(x, y_0) - h'(x). \end{aligned}$$

Thus we obtain

$$h'(x) = -\frac{\partial u}{\partial y}(x, y_0).$$

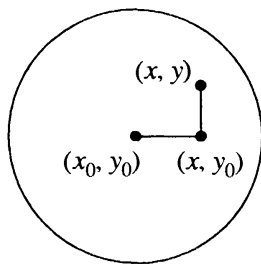
This has the solution

$$h(x) = - \int_{x_0}^x \frac{\partial u}{\partial y}(s, y_0) ds + C,$$

where C is a genuine constant. Thus we see that a harmonic conjugate $v(x, y)$ for $u(x, y)$ is given explicitly by

$$(5.3) \quad v(x, y) = \int_{y_0}^y \frac{\partial u}{\partial x}(x, t) dt - \int_{x_0}^x \frac{\partial u}{\partial y}(s, y_0) ds + C.$$

The formula (5.3) is also valid if D is the entire complex plane, or if D is an open disk with center (x_0, y_0) . Note that if we specify $v(x_0, y_0) = 0$, then $C = 0$, and the solution is unique.



We summarize in the following theorem.

Theorem. *Let D be an open disk, or an open rectangle with sides parallel to the axes, and let $u(x, y)$ be a harmonic function on D . Then there is a harmonic function $v(x, y)$ on D such that $u + iv$ is analytic on D . The harmonic conjugate v is unique, up to adding a constant.*

We will see in Chapter III that this theorem holds in star-shaped domains. However, the theorem fails in annuli and in the punctured plane (Exercise 7). Roughly speaking, the theorem holds only in domains that have no “holes.” Such domains are called “simply connected domains.” They will be discussed in Chapter VIII.

Exercises for II.5

1. Show that the following functions are harmonic, and find harmonic conjugates:

(a) $x^2 - y^2$	(c) $\sinh x \sin y$	(e) $\tan^{-1}(y/x), \quad x > 0$
(b) $xy + 3x^2y - y^3$	(d) $e^{x^2-y^2} \cos(2xy)$	(f) $x/(x^2 + y^2)$
2. Show that if v is a harmonic conjugate for u , then $-u$ is a harmonic conjugate for v .
3. Define $u(z) = \operatorname{Im}(1/z^2)$ for $z \neq 0$, and set $u(0) = 0$.
 - (a) Show that all partial derivatives of u with respect to x exist at all points of the plane $\mathbb{C}$, as do all partial derivative of u with respect to y .
 - (b) Show that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$.
 - (c) Show that u is *not* harmonic on $\mathbb{C}$.
 - (d) Show that $\frac{\partial^2 u}{\partial x \partial y}$ does not exist at $(0, 0)$.
4. Show that if $h(z)$ is a complex-valued harmonic function (solution of Laplace's equation) such that $zh(z)$ is also harmonic, then $h(z)$ is analytic.
5. Show that Laplace's equation in polar coordinates is

$$\frac{\partial^2 u}{\partial r^2} + \frac{1}{r} \frac{\partial u}{\partial r} + \frac{1}{r^2} \frac{\partial^2 u}{\partial \theta^2} = 0.$$

6. Show using Laplace's equation in polar coordinates that $\log|z|$ is harmonic on the punctured plane $\mathbb{C} \setminus \{0\}$.
7. Show that $\log|z|$ has no conjugate harmonic function on the punctured plane $\mathbb{C} \setminus \{0\}$, though it does have a conjugate harmonic function on the slit plane $\mathbb{C} \setminus (-\infty, 0]$.
8. Show using Laplace's equation in polar coordinates that $u(re^{i\theta}) = \theta \log r$ is harmonic. Use the polar form of the Cauchy-Riemann equations (Exercise 3.8) to find a harmonic conjugate v for u . What is the analytic function $u + iv$?

6. Conformal Mappings

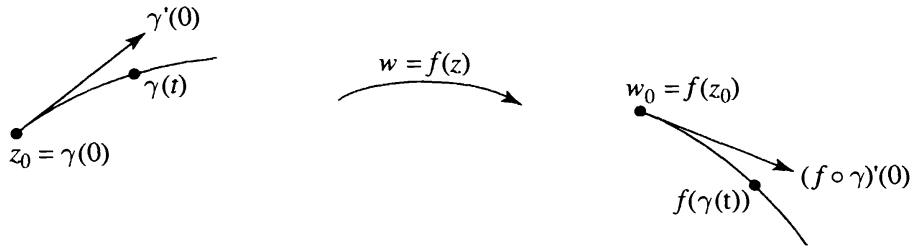
Let $\gamma(t) = x(t) + iy(t)$, $0 \leq t \leq 1$, be a smooth parameterized curve terminating at $z_0 = \gamma(0)$. We refer to

$$\gamma'(0) = \lim_{t \rightarrow 0} \frac{\gamma(t) - \gamma(0)}{t} = x'(0) + iy'(0)$$

as the **tangent vector** to the curve γ at z_0 . It is the complex representation of the usual tangent vector. We define the **angle between two curves** at z_0 to be the angle between their tangent vectors at z_0 .

Theorem. *If $\gamma(t)$, $0 \leq t \leq 1$, is a smooth parameterized curve terminating at $z_0 = \gamma(0)$, and $f(z)$ is analytic at z_0 , then the tangent to the curve $f(\gamma(t))$ terminating at $f(z_0)$ is*

$$(6.1) \quad (f \circ \gamma)'(0) = f'(z_0)\gamma'(0).$$



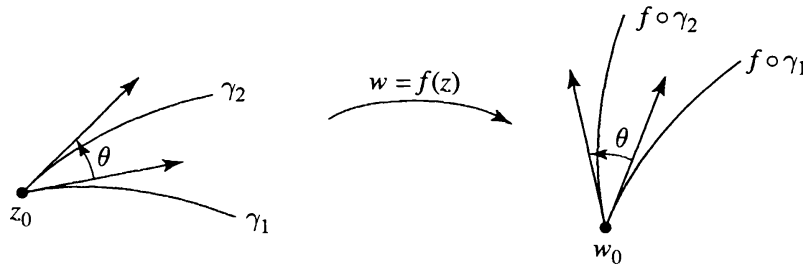
The proof is a close relative of the proof of the chain rule for the composition of analytic functions (Section 2). If $\gamma'(0) \neq 0$, then $\gamma(t) \neq \gamma(0)$ for t near 0, $t \neq 0$, so we may write

$$\frac{f(\gamma(t)) - f(\gamma(0))}{t} = \frac{f(\gamma(t)) - f(\gamma(0))}{\gamma(t) - \gamma(0)} \frac{\gamma(t) - \gamma(0)}{t}$$

and pass to the limit, to obtain the formula (6.1). If $\gamma'(0) = 0$, then proceeding as in Section 2, we obtain $(f \circ \gamma)'(0) = 0$, and again the formula holds.

We may think of the tangent vector as a vector in the plane with tail at z_0 . Composing a parameterized curve with $f(z)$ then has the effect upon the tangent vector of multiplying it by $f'(z_0)$ (complex multiplication) and moving the tail to $w_0 = f(z_0)$. If the tangent vector is represented by $z - z_0$, then the tangent to the image curve is represented by $w - f(z_0) = f'(z_0)(z - z_0)$. As far as the tangent vector at z_0 is concerned, the effect of composing with $f(z)$ is the same as the effect of composing with the function $f(z_0) + f'(z_0)(z - z_0)$, which is the first-order Taylor approximation to $f(z)$ at z_0 . The remainder term $R(z)$ in the Taylor approximation satisfies $R(z)/(z - z_0) \rightarrow 0$ as $z \rightarrow z_0$, so that $R(z)$ has no effect on tangent vectors.

A function is **conformal** if it preserves angles. More precisely, we say that a smooth complex-valued function $g(z)$ is **conformal at z_0** if whenever γ_0 and γ_1 are two curves terminating at z_0 with nonzero tangents, then the curves $g \circ \gamma_0$ and $g \circ \gamma_1$ have nonzero tangents at $g(z_0)$ and the angle from $(g \circ \gamma_0)'(z_0)$ to $(g \circ \gamma_1)'(z_0)$ is the same as the angle from $\gamma_0'(z_0)$ to $\gamma_1'(z_0)$. A **conformal mapping** of one domain D onto another V is a continuously differentiable function that is conformal at each point of D and that maps D one-to-one onto V .



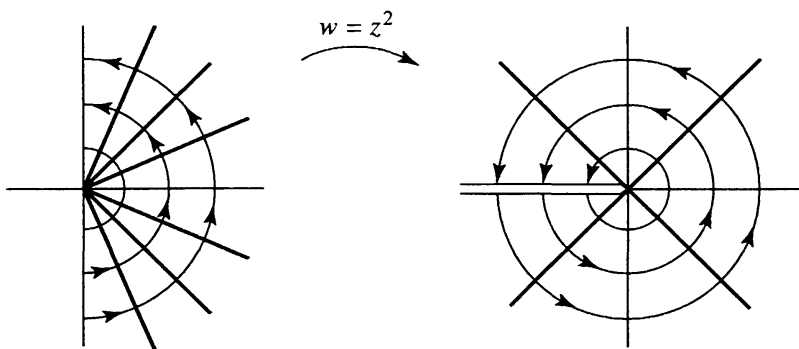
The translation $f(z) = z + b$ and the complex multiplication $g(z) = az$, where $a \neq 0$, evidently preserve angles, hence are conformal everywhere. They are conformal mappings of the complex plane onto itself. On the other hand, the function $a\bar{z}$ reverses angles and orientation, so it is not conformal. For $n > 1$, the function z^n multiplies angles at the origin by n , so it is not conformal at $z = 0$. The following theorem shows that z^n is conformal at any point z other than 0.

Theorem. *If $f(z)$ is analytic at z_0 and $f'(z_0) \neq 0$, then $f(z)$ is conformal at z_0 .*

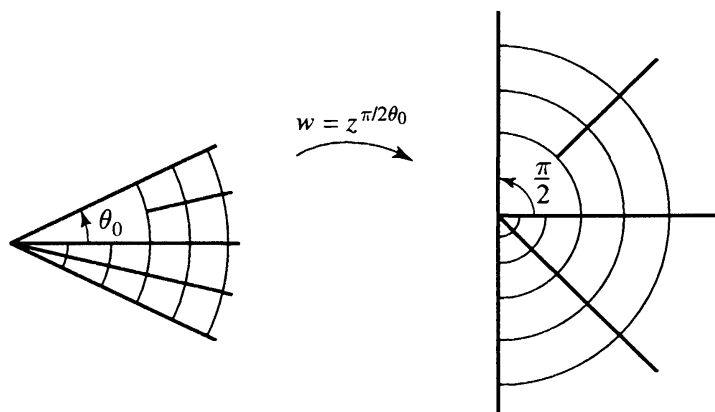
Let γ_0 and γ_1 be two curves terminating at z_0 with nonzero tangents. By the preceding theorem, the tangents to the curves $g \circ \gamma_0$ and $g \circ \gamma_1$ are obtained by multiplying the respective tangents to γ_0 and γ_1 by $f'(z_0)$. Thus the arguments of both tangents are increased by the same angle, namely the argument of $f'(z_0)$. Consequently, the angle between them is preserved.

There is a converse to this theorem, to the effect that conformal mappings are analytic. Though the result is elementary, we postpone it to Section IV.8. (But see Exercise 9.)

Example. The function $w = z^2$ maps the right half-plane $\{\operatorname{Re} z > 0\}$ conformally onto the slit plane $\mathbb{C} \setminus (-\infty, 0]$. For any fixed θ_0 , $0 < \theta_0 \leq \pi/2$, it maps the sector $\{|\arg z| < \theta_0\}$ conformally onto the sector $\{|\arg z| < 2\theta_0\}$ of twice the aperture.



Example. Fix θ_0 , $0 < \theta_0 \leq \pi$. If $0 < a < \pi/\theta_0$, the function z^a maps the sector $\{|\arg z| < \theta_0\}$ conformally onto the sector $\{|\arg z| < a\theta_0\}$. In particular, the function $z^{\pi/2\theta_0}$ maps the sector $\{|\arg z| < \theta_0\}$ conformally onto the right half-plane.



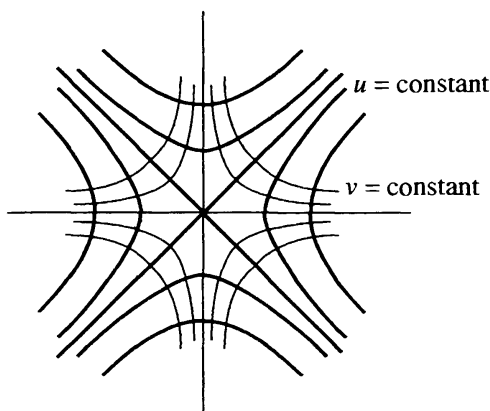
Example. The exponential function e^z is conformal at each point $z \in \mathbb{C}$, since its derivative does not vanish at z . Its image is the punctured plane $\mathbb{C} \setminus \{0\}$. However, it is not a conformal mapping of the plane onto the punctured plane, since it is not one-to-one. Its restriction to the horizontal strip $\{|\operatorname{Im} z| < \pi\}$ is a conformal mapping of the strip onto the slit plane $\mathbb{C} \setminus (-\infty, 0]$.

Example. The principal branch $\operatorname{Log} z$ of the logarithm is a conformal mapping of the slit plane $\mathbb{C} \setminus (-\infty, 0]$ onto the horizontal strip $\{|\operatorname{Im} w| < \pi\}$. See the figure in Section I.6.

Any conformal mapping carries orthogonal curves to orthogonal curves, and it carries orthogonal families of curves to orthogonal families of curves. In the case of the exponential function e^z , the orthogonal grid consisting of horizontal and vertical lines is mapped to an orthogonal grid consisting of rays emanating from the origin and circles centered at the origin.

Something similar happens for any nonconstant analytic function $f = u + iv$ on a domain D . Fix a point z_0 where $f'(z_0) \neq 0$, and consider the two curves $\{u(z) = u(z_0)\}$ and $\{v(z) = v(z_0)\}$, which meet at z_0 . The function $f(z)$ is one-to-one near z_0 , it maps the part of the level set $\{u(z) = u(z_0)\}$ near z_0 to a vertical line segment through $f(z_0)$, and it maps the part of the level set $\{v(z) = v(z_0)\}$ near z_0 to a horizontal line segment through $f(z_0)$. Since these line segments are orthogonal at $f(z_0)$, the level sets of u and v are orthogonal at z_0 . Thus the two families of curves $\{u = \text{constant}\}$ and $\{v = \text{constant}\}$ are orthogonal except at points where $f'(z_0) = 0$.

Example. For $f(z) = z^2 = x^2 - y^2 + 2ixy$, the families of curves $u = \text{constant}$ and $v = \text{constant}$ form two families of hyperbolas that are orthogonal except at the origin.



Exercises for II.6

1. Sketch the families of level curves of u and v for the following functions $f = u + iv$. (a) $f(z) = 1/z$, (b) $f(z) = 1/z^2$, (c) $f(z) = z^6$. Determine where $f(z)$ is conformal and where it is not conformal.
2. Sketch the families of level curves of u and v for $f(z) = \text{Log } z = u + iv$. Relate your sketch to one of the figures in this section.
3. Sketch the families of level curves of u and v for the functions $f = u + iv$ given by (a) $f(z) = e^z$, (b) $f(z) = e^{\alpha z}$, where α is complex. Determine where $f(z)$ is conformal and where it is not conformal.

4. Find a conformal map of the horizontal strip $\{-A < \operatorname{Im} z < A\}$ onto the right half-plane $\{\operatorname{Re} w > 0\}$. *Hint.* Recall the discussion of the exponential function, or refer to the preceding problem.
5. Find a conformal map of the wedge $\{-B < \arg z < B\}$ onto the right half-plane $\{\operatorname{Re} w > 0\}$. Assume $0 < B < \pi$.
6. Determine where the function $f(z) = z + 1/z$ is conformal and where it is not conformal. Show that for each w , there are at most two values z for which $f(z) = w$. Show that if $r > 1$, $f(z)$ maps the circle $\{|z| = r\}$ onto an ellipse, and that $f(z)$ maps the circle $\{|z| = 1/r\}$ onto the same ellipse. Show that $f(z)$ is one-to-one on the exterior domain $D = \{|z| > 1\}$. Determine the image of D under $f(z)$. Sketch the images under $f(z)$ of the circles $\{|z| = r\}$ for $r > 1$, and sketch also the images of the parts of the rays $\{\arg z = \beta\}$ lying in D .
7. For the function $f(z) = z + 1/z = u + iv$, sketch the families of level curves of u and v . Determine the images under $f(z)$ of the top half of the unit disk, the bottom half of the unit disk, the part of the upper half-plane outside the unit disk, and the part of the lower half-plane outside the unit disk. *Hint.* Start by locating the images of the curves where $u = 0$, where $v = 0$, and where $v = 1$. Note that the level curves are symmetric with respect to the real and imaginary axes, and they are invariant under the inversion $z \mapsto 1/z$ in the unit circle.
8. Consider $f(z) = z + e^{i\alpha}/z$, where $0 < \alpha < \pi$. Determine where $f(z)$ is conformal and where it is not conformal. Sketch the images under $f(z)$ of the unit circle $\{|z| = 1\}$ and the intervals $(-\infty, -1]$ and $[+1, +\infty)$ on the real axis. Show that $w = f(z)$ maps $\{|z| > 1\}$ conformally onto the complement of a slit in the w -plane. Sketch roughly the images of the segments of rays outside the unit circle $\{\arg z = \beta, |z| \geq 1\}$ under $f(z)$. At what angles do they meet the slit, and at what angles do they approach ∞ ?
9. Let $f = u + iv$ be a continuously differentiable complex-valued function on a domain D such that the Jacobian matrix of f does not vanish at any point of D . Show that if f maps orthogonal curves to orthogonal curves, then either f or $\bar{f}$ is analytic, with nonvanishing derivative.

7. Fractional Linear Transformations

A **fractional linear transformation** is a function of the form

$$(7.1) \quad w = f(z) = \frac{az + b}{cz + d},$$

where a, b, c, d are complex constants satisfying $ad - bc \neq 0$. Fractional linear transformations are also called **Möbius transformations**. Since

$$f'(z) = \frac{ad - bc}{(cz + d)^2},$$

the condition $ad - bc \neq 0$ simply guarantees that $f(z)$ is not constant.

If we multiply each of the parameters a, b, c, d in (7.1) by the same nonzero constant, we obtain the same function. Thus different choices of the parameters may lead to the same fractional linear transformation.

Example. A function of the form $f(z) = az + b$, where $a \neq 0$, is called an **affine transformation**. These are the fractional linear transformations of the form (7.1) with $c = 0$. Special cases are the **translations** $z \mapsto z + b$ and the **dilations** $z \mapsto az$.

Example. The fractional linear transformation $f(z) = 1/z$ is called an **inversion**.

It is convenient to regard a fractional linear transformation as a map from the extended complex plane $\mathbb{C}^* = \mathbb{C} \cup \{\infty\}$ to itself. If $f(z)$ is affine, we define $f(\infty) = \infty$. Otherwise, $f(z)$ has the form (7.1) where $c \neq 0$, and we define $f(-d/c) = \infty$ and

$$f(\infty) = \lim_{z \rightarrow \infty} f(z) = \lim_{z \rightarrow \infty} \frac{a + b/z}{c + d/z} = \frac{a}{c}.$$

Thus translations and dilations map ∞ to ∞ , while the inversion $z \mapsto 1/z$ interchanges 0 and ∞ .

The inverse of a fractional linear transformation is again a fractional linear transformation. To see this, we solve (7.1) for z , to obtain

$$z = \frac{-dw + b}{cw - a}.$$

The condition on the coefficients is satisfied, since $(-d)(-a) - bc = ad - bc \neq 0$, or alternatively, since the function $z = z(w)$ is not constant. This shows that each fractional linear transformation is a one-to-one function from the extended complex plane onto itself.

The composition of two fractional linear transformations is again a fractional linear transformation. To see this, suppose $f(z) = (az + b)/(cz + d)$ and $g(z) = (\alpha z + \beta)/(\gamma z + \delta)$, and compute

$$f(g(z)) = \frac{a((\alpha z + \beta)/(\gamma z + \delta)) + b}{c((\alpha z + \beta)/(\gamma z + \delta)) + d} = \frac{(a\alpha + b\gamma)z + a\beta + b\delta}{(c\alpha + d\gamma)z + c\beta + d\delta}.$$

Since the composition $f \circ g$ cannot be constant, the condition on the parameters is met, and $f \circ g$ is a fractional linear transformation.

Note that the composition corresponds to matrix multiplication,

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix} = \begin{pmatrix} a\alpha + b\gamma & a\beta + b\delta \\ c\alpha + d\gamma & c\beta + d\delta \end{pmatrix}.$$

The condition $ad - bc \neq 0$ on the parameters is simply the condition that the matrix associated with the fractional linear transformation has nonzero determinant, that is, that the matrix is invertible.

The fact that matrix multiplication corresponds to composition can be reformulated in the language of group theory. If we assign to each 2×2 invertible matrix the corresponding fractional linear transformation, we obtain what is called a “group homomorphism,” from the group of 2×2 invertible matrices with complex entries onto the group of fractional linear transformations with operation composition.

A fractional linear transformation depends on four complex parameters. One of these can be adjusted without changing the transformation, for instance by multiplying all the parameters by the same nonzero constant. That leaves three parameters to be specified. The next theorem shows that there are three independent complex parameters that describe fractional linear transformations uniquely, namely, the images of any three prescribed points.

Theorem. *Given any three distinct points z_0, z_1, z_2 in the extended complex plane, and given any three distinct values w_0, w_1, w_2 in the extended complex plane, there is a unique fractional linear transformation $w = w(z)$ such that $w(z_0) = w_0$, $w(z_1) = w_1$, and $w(z_2) = w_2$.*

To establish the existence assertion, it suffices to show that any three distinct points can be mapped by a fractional linear transformation to 0, 1, and ∞ . Indeed, if f maps z_0, z_1, z_2 respectively to 0, 1, ∞ , and g maps w_0, w_1, w_2 respectively to 0, 1, ∞ , then the composition $g^{-1} \circ f$, of f followed by the inverse of g , maps z_0, z_1, z_2 to w_0, w_1, w_2 . If now none of the points z_0, z_1, z_2 is ∞ , a transformation mapping them to 0, 1, ∞ is given explicitly by

$$(7.2) \quad w = f(z) = \frac{z - z_0}{z - z_2} \frac{z_1 - z_2}{z_1 - z_0}.$$

If one of the z_j 's is ∞ , we define $f(z)$ by sending that z_j to ∞ in the above formula. For instance, if $z_0 = \infty$, we rewrite the right-hand side of (7.2) as

$$\frac{(z/z_0) - 1}{z - z_2} \frac{z_1 - z_2}{(z_1/z_0) - 1}$$

and take a limit as $z_0 \rightarrow \infty$, to obtain

$$w = f(z) = \frac{z_1 - z_2}{z - z_2}.$$

This maps ∞, z_1, z_2 to $0, 1, \infty$. There are similar formulae for the cases $z_1 = \infty$ and $z_2 = \infty$.

For the uniqueness, suppose first that $f(z)$ is a fractional linear transformation that fixes $0, 1$, and ∞ . Since $f(\infty) = \infty$, $f(z) = az + b$ for some $a \neq 0$. From $f(0) = 0$ we obtain $b = 0$, and from $f(1) = 1$ we obtain $a = 1$. Thus $f(z) = z$ is the identity transformation.

Now suppose that $g(z)$ and $h(z)$ are both fractional linear transformations mapping the z_j 's to the respective w_j 's. Let $k(z)$ map the z_j 's respectively to $0, 1, \infty$. Then $f = k \circ h^{-1} \circ g \circ k^{-1}$ maps 0 to 0 , 1 to 1 , and ∞ to ∞ . Hence $f(z) = z$ is the identity, and $g = h \circ k^{-1} \circ f \circ k = h \circ k^{-1} \circ k = h$. This establishes the uniqueness assertion of the theorem.

Exercise. Find the fractional linear transformation mapping -1 to 0 , ∞ to 1 , and i to ∞ .

Solution. We could use (7.2) and send z_1 to ∞ . However, it is easier to proceed directly. Since $w(i) = \infty$, we place $z - i$ in the denominator, and since $w(-1) = 0$, we place $z + 1$ in the numerator, to obtain $w(z) = a(z + 1)/(z - i)$. Since $w(z) \rightarrow 1$ as $z \rightarrow \infty$, we obtain $a = 1$, and hence $w(z) = (z + 1)/(z - i)$.

Theorem. *Every fractional linear transformation is a composition of dilations, translations, and inversions.*

A fractional linear transformation mapping ∞ to ∞ has the form $w = az + b$ where $a \neq 0$. This is the composition of the translation $z \mapsto z + b/a$ and the dilation $z \mapsto az$:

$$z \mapsto z + b/a \mapsto a(z + b/a) = az + b.$$

If $w(\infty)$ is finite, then w has the form (7.1) where $c \neq 0$. In this case we may divide each of the parameter values by c and assume that $c = 1$. Then $w(z) = (az + b)/(z + d)$. Now we conjure up by magic the identity

$$w(z) = \frac{az + b}{z + d} = a + \frac{b - ad}{z + d}.$$

This expression allows us to represent $w(z)$ as

$$z \mapsto z + d \mapsto \frac{1}{z + d} \mapsto \frac{b - ad}{z + d} \mapsto a + \frac{b - ad}{z + d},$$

and consequently, $w(z)$ is a composition of a translation, an inversion, a dilation, and a translation.

Theorem. *A fractional linear transformation maps circles in the extended complex plane to circles.*

It suffices to establish the theorem for translations, dilations, and inversions, since every fractional linear transformation is a composition of these.

It is clear (or it should be) that translations and dilations map circles to circles. Thus it suffices to check that the inversion $w = 1/z$ maps circles to circles.

Consider first a circle that does not pass through ∞ . It has an equation of the form $|z - a|^2 = r^2$, where a is its center and r its radius. The image of the circle under the inversion $w = 1/z$ consists of points satisfying $|1 - aw|^2 = r^2|w|^2$, that is,

$$\begin{aligned} 0 &= |1 - aw|^2 - r^2|w|^2 = (1 - aw)(\overline{1 - aw}) - r^2|w|^2 \\ &= (|a|^2 - r^2)|w|^2 - aw - \overline{aw} + 1. \end{aligned}$$

Set $w = u + iv$ where u and v are real. The equation assumes the form

$$(|a|^2 - r^2)(u^2 + v^2) + Au + Bv + 1 = 0,$$

where A and B are real constants. If $r = |a|$, the equation represents a straight line in the plane, which is a circle through ∞ . If $r \neq |a|$, this is a quadratic equation in u and v of the form met in Section I.3. It has more than one solution, so the solutions form a circle. (To obtain the center and radius, complete the square.)

Next consider a circle passing through ∞ , that is, a straight line. It has an equation of the form $Cx + Dy = E$. A calculation similar to the one given above shows that the image is a circle if $E \neq 0$ and a straight line if $E = 0$ (Exercise 6).

Exercise. Find the equation of the fractional linear transformation mapping 0 to -1 , i to 0, and ∞ to 1.

Solution. Since $i \rightarrow 0$, we can normalize a to be 1 and write the fractional linear transformation in the form $w = (z - i)/(cz + d)$. The condition $0 \mapsto -1$ yields $-i/d = -1$, and so $d = i$. Finally, the condition $\infty \mapsto 1$ yields $1/c = 1$, and so $c = 1$. Thus $w = (z - i)/(z + i)$.

Exercise. Determine the images of each of the following sets under the above fractional linear transformation: (a) the imaginary axis, (b) the right half-plane, (c) the real axis, (d) the upper half-plane, (e) the horizontal line through i . Sketch the images of horizontal lines and of vertical lines under the transformation.

Solution. We will solve this exercise without referring to the explicit formula for the transformation. We use two facts. First, to determine the image of a circle under a fractional linear transformation, it suffices to determine the images of three points on the circle. Since three points determine a circle, the image of the circle is then the circle passing through the three image points. Second, fractional linear transformations map orthogonal circles to orthogonal circles, since they are conformal.

(a) The three points 0, i , ∞ lie on the circle corresponding to the imaginary axis in the extended complex plane. The image of the imaginary axis is then the circle through the three image points $-1, 0, +1$, which is the

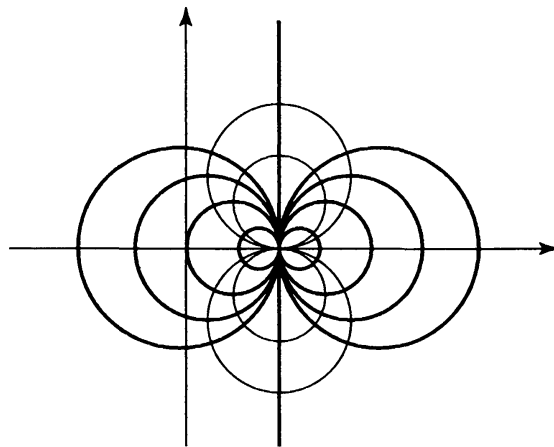
real line.

(b) The ordered triple $0, i, \infty$ is mapped to the ordered triple $-1, 0, 1$, so the image w moves forward on the real line as z moves upwards on the imaginary axis. Since orientations are preserved, the right half-plane is mapped to the domain on the right of the positively traversed real line, which is the lower half-plane.

(c) The real axis is mapped to a circle through $w(0) = -1$ and $w(\infty) = +1$. Since the real and imaginary axes are orthogonal, their images are orthogonal. Thus the image of the real line is a circle through ± 1 that is orthogonal to the real line. There is only one such circle, the unit circle. Consequently, the image of the real line is the unit circle $\{|w| = 1\}$.

(d) Since the image of the real line is the unit circle, the image of the upper half-plane does not cross the unit circle, and it must coincide either with the inside $\{|w| < 1\}$ or with the exterior domain $\{|w| > 1\}$ together with ∞ . Since i is mapped to 0 , which is inside the unit circle, the image of the upper half-plane is the inside, that is, it is the open unit disk $\{|w| < 1\}$.

(e) The image of the horizontal line through i is a circle passing through 0 and 1 , and it lies inside the unit disk, so it must be the circle centered at $\frac{1}{2}$ of radius $\frac{1}{2}$.



The image of any horizontal line is a circle through $w(\infty) = 1$, and it is orthogonal to the real line (the image of the imaginary axis). These images of the horizontal lines form a pencil of circles as sketched in the figure. The images of vertical lines are circles through $w(\infty) = 1$. Since the real axis is the image of the imaginary axis, these circles must be tangent to the real axis at 1 . The images of the vertical lines are also sketched in the figure. Note that the images of the horizontal and vertical lines are orthogonal to each other.

Exercises for II.7

1. Compute explicitly the fractional linear transformations determined by the following correspondences of triples:

- | | |
|---|---|
| (a) $(1 + i, 2, 0) \mapsto (0, \infty, i - 1)$ | (e) $(1, 2, \infty) \mapsto (0, 1, \infty)$ |
| (b) $(0, 1, \infty) \mapsto (1, 1 + i, 2)$ | (f) $(0, \infty, i) \mapsto (0, 1, \infty)$ |
| (c) $(\infty, 1 + i, 2) \mapsto (0, 1, \infty)$ | (g) $(0, 1, \infty) \mapsto (0, \infty, i)$ |
| (d) $(-2, i, 2) \mapsto (1 - 2i, 0, 1 + 2i)$ | (h) $(1, i, -1) \mapsto (1, 0, -1)$ |

- Consider the fractional linear transformation in Exercise 1a above, which maps $1 + i$ to 0 , 2 to ∞ , and 0 to $i - 1$. Without referring to an explicit formula, determine the image of the circle $\{|z - 1| = 1\}$, the image of the disk $\{|z - 1| < 1\}$, and the image of the real axis.
- Consider the fractional linear transformation that maps 1 to i , 0 to $1 + i$, and -1 to 1 . Determine the image of the unit circle $\{|z| = 1\}$, the image of the open unit disk $\{|z| < 1\}$, and the image of the imaginary axis. Illustrate with a sketch.
- Consider the fractional linear transformation that maps -1 to $-i$, 1 to $2i$, and i to 0 . Determine the image of the unit circle $\{|z| = 1\}$, the image of the open unit disk $\{|z| < 1\}$, and the image of the interval $[-1, +1]$ on the real axis. Illustrate with a sketch.
- What is the image of the horizontal line through i under the fractional linear transformation that interchanges 0 and 1 and maps -1 to $1 + i$? Illustrate with a sketch.
- Show that the image of a straight line under the inversion $z \mapsto 1/z$ is a straight line or circle, depending on whether the line passes through the origin.
- Show that the fractional linear transformation $f(z) = (az + b)/(cz + d)$ is the identity mapping z if and only if $b = c = 0$ and $a = d \neq 0$.
- Show that any fractional linear transformation can be represented in the form $f(z) = (az + b)/(cz + d)$, where $ad - bc = 1$. Is this representation unique?
- Show that the fractional linear transformations that are real on the real axis are precisely those that can be expressed in the form $(az + b)/(cz + d)$, where a, b, c , and d are real.
- Suppose the fractional linear transformation $(az + b)/(cz + d)$ maps $\mathbb{R}$ to $\mathbb{R}$, and $ad - bc = 1$. Show that a, b, c , and d are real or they are all pure imaginary.
- Two maps f and g are **conjugate** if there is h such that $g = h \circ f \circ h^{-1}$. Here the conjugating map h is assumed to be one-to-one, with appropriate domain and range. We can think of f and g as the “same” map, after the change of variable $w = h(z)$. A point z_0 is a **fixed point** of f if $f(z_0) = z_0$. Show the following. (a) If f

is conjugate to g , then g is conjugate to f . (b) If f_1 is conjugate to f_2 and f_2 to f_3 , then f_1 is conjugate to f_3 . (c) If f is conjugate to g , then $f \circ f$ is conjugate to $g \circ g$, and more generally, the m -fold composition $f \circ \cdots \circ f$ (m times) is conjugate to $g \circ \cdots \circ g$ (m times). (d) If f and g are conjugate, then the conjugating function h maps the fixed points of f to the fixed points of g . In particular, f and g have the same number of fixed points.

12. Classify the conjugacy classes of fractional linear transformations by establishing the following:

- (a) A fractional linear transformation that is not the identity has either 1 or 2 fixed points, that is, points satisfying $f(z_0) = z_0$.
- (b) If a fractional linear transformation $f(z)$ has two fixed points, then it is conjugate to the dilation $z \mapsto az$ with $a \neq 0$, $a \neq 1$, that is, there is a fractional linear transformation $h(z)$ such that $h(f(z)) = ah(z)$. Is a unique? *Hint.* Consider a fractional linear transformation that maps the fixed points to 0 and ∞ .
- (c) If a fractional linear transformation $f(z)$ has exactly one fixed point, then it is conjugate to the translation $\zeta \mapsto \zeta + 1$. In other words, there is a fractional linear transformation $h(z)$ such that $h(f(h^{-1}(\zeta))) = \zeta + 1$, or equivalently, such that $h(f(z)) = h(z) + 1$. *Hint.* Consider a fractional linear transformation that maps the fixed point to ∞ .

III

Line Integrals and Harmonic Functions

In Sections 1 and 2 we review multivariable integral calculus in order to prepare for complex integration in the next chapter. The salient features are Green's theorem and independence of path for line integrals. In Section 3 we introduce harmonic functions, and in Sections 4 and 5 we discuss the mean value property and the maximum principle for harmonic functions. Sections 6 and 7 include various applications to physics. The student may proceed directly to complex integration in the next chapter after paging through the review of multivariable calculus in Sections 1 and 2 and reading about harmonic conjugates in Section 3.

1. Line Integrals and Green's Theorem

Line integrals play an important role in complex analysis. In this section and the next we review line integrals in the plane, without filling in all the details. We begin by saying something about paths and curves.

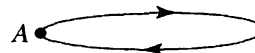
A **path** in the plane from A to B is a continuous function $t \mapsto \gamma(t)$ on some parameter interval $a \leq t \leq b$ such that $\gamma(a) = A$ and $\gamma(b) = B$. The path is **simple** if $\gamma(s) \neq \gamma(t)$ when $s \neq t$. The path is **closed** if it starts and ends at the same point, that is, $\gamma(a) = \gamma(b)$. A **simple closed path** is a closed path γ such that $\gamma(s) \neq \gamma(t)$ for $a \leq s < t < b$.



simple path



path (not simple)



simple closed path

If $\gamma(t)$, $a \leq t \leq b$, is a path from A to B , and if $\phi(s)$, $\alpha \leq s \leq \beta$, is a strictly increasing continuous function satisfying $\phi(\alpha) = a$ and $\phi(\beta) = b$, then the composition $\gamma(\phi(s))$, $\alpha \leq s \leq \beta$, is also a path from A to B . The composition $\gamma \circ \phi$ is a “reparametrization” of γ . For our purposes we

can usually regard γ and any of its reparametrizations as being the same path. (Technically, we should consider equivalence classes of paths.) Note that reparametrization preserves the order of points of a path; that is, it preserves orientation.

The **trace** of the path γ is its image $\gamma([a, b])$, which is a subset of the plane. When it is clear from context, we will denote the trace of a path also by γ . It will not be until Chapter VIII that we need to be careful about distinguishing the path γ from its trace.

If one path ends where another begins, the two paths can be concatenated by following one and then the other, after suitable reparametrization.

A **smooth path** is a path that can be represented in the form $\gamma(t) = (x(t), y(t))$, $a \leq t \leq b$, where the functions $x(t)$ and $y(t)$ are smooth, that is, have as many derivatives as is necessary for whatever is being asserted to be true. A **piecewise smooth path** is a concatenation of smooth paths. By a **curve** we mean (usually) a smooth or piecewise smooth path.

Let γ be a path in the plane from A to B , and let $P(x, y)$ and $Q(x, y)$ be continuous complex-valued functions on γ . We consider successive points on the path, $A = (x_0, y_0)$, (x_1, y_1) , $\dots$, $B = (x_n, y_n)$, and we form the sum

$$(1.1) \quad \sum P(x_j, y_j)(x_{j+1} - x_j) + \sum Q(x_j, y_j)(y_{j+1} - y_j).$$

If these sums have a limit as the distances between the successive points on γ tend to 0, we define the limit to be the **line integral of $Pdx + Qdy$ along γ** , and we denote it by

$$(1.2) \quad \int_{\gamma} P dx + Q dy.$$

Suppose the path $\gamma(t) = (x(t), y(t))$, $a \leq t \leq b$, is continuously differentiable, that is, the parameter functions $x(t)$ and $y(t)$ are continuously differentiable. Suppose the parameter values t_j satisfy $x(t_j) = x_j$, $y(t_j) = y_j$, where $a = t_0 < t_1 < \dots < t_n = b$. By the mean value theorem, there are points t_j^* between t_j and t_{j+1} such that $x(t_{j+1}) - x(t_j) = x'(t_j^*)(t_{j+1} - t_j)$. If we substitute this into the first sum in (1.1), we obtain

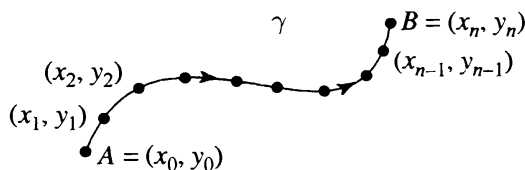
$$\sum P(x(t_j), y(t_j))x'(t_j^*)(t_{j+1} - t_j),$$

which is a Riemann sum approximating the integral $\int_a^b P(x(t), y(t))x'(t)dt$. Similarly, the second sum in (1.1) is a Riemann sum approximating the integral $\int_a^b Q(x(t), y(t))y'(t)dt$. As the distances between the successive t_j 's tend to 0, the sums in (1.1) converge to an ordinary garden-variety Riemann integral, and we obtain

$$(1.3) \quad \int_{\gamma} P dx + Q dy = \int_a^b P(x(t), y(t)) \frac{dx}{dt} dt + \int_a^b Q(x(t), y(t)) \frac{dy}{dt} dt.$$

Thus to evaluate a line integral over a smooth curve, we simply parametrize the curve by $t \mapsto (x(t), y(t))$, calculate the derivatives dx/dt and dy/dt of

the components, and plug these into the definite integral in (1.3). To evaluate the line integral over a path that is only piecewise smooth, we parametrize each smooth subpath, calculate the corresponding integrals by (1.3), and add them.



Note that the definition of the line integral over γ is independent of the parameterization of γ . The parameterization enters only in determining the ordering of the points on the curve γ . Thus different parameterizations give the same integral in (1.3). Also note that if we reverse the direction of γ , then the line integral is replaced by its negative.

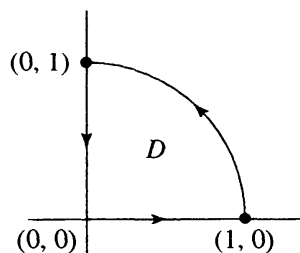
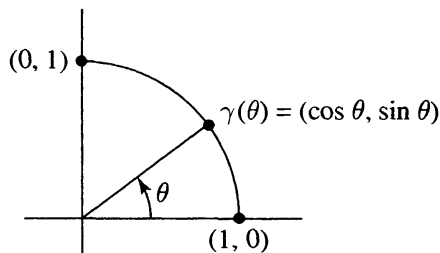
Example. To evaluate $\int_{\gamma} xy \, dx$, where γ is the quarter-circle from $(1, 0)$ to $(0, 1)$ on the unit circle, we parametrize γ by

$$(x(\theta), y(\theta)) = (\cos \theta, \sin \theta), \quad 0 \leq \theta \leq \pi/2,$$

and we substitute into (1.3). This gives

$$\begin{aligned} \int_{\gamma} xy \, dx &= \int_0^{\pi/2} \cos \theta \sin \theta \, d(\cos \theta) = - \int_0^{\pi/2} \cos \theta \sin^2 \theta \, d\theta \\ &= - \frac{\sin^3 \theta}{3} \Big|_0^{\pi/2} = -\frac{1}{3}. \end{aligned}$$

Note that the sign is correct, since $xy \geq 0$ in the curve γ , while $dx < 0$ on the curve (since x decreases on the curve).



A domain D has **piecewise smooth boundary** if the boundary of D can be decomposed into a finite number of smooth curves meeting only at endpoints. By “smooth” we usually mean “continuously differentiable,” though in applications the curves making up the boundary will usually be straight line segments or arcs of circles. We denote the boundary of D by ∂D . For purposes of integration, the **orientation** of ∂D is chosen so that D lies on the left of a curve in ∂D as we traverse the boundary curve in the positive direction, that is, as the parameter value increases.

Example. To evaluate $\int_{\partial D} xy \, dx$, where D is the quarter-disk in the first quadrant, we divide the integral into three pieces,

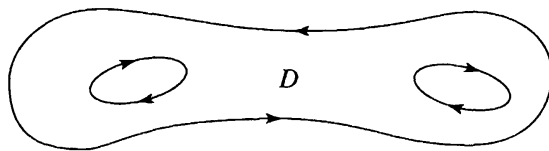
$$\int_{\partial D} xy \, dx = \int_{\gamma} xy \, dx + \int_{(1,0)}^{(0,0)} xy \, dx + \int_{(0,0)}^{(1,0)} xy \, dx,$$

where γ is the quarter-circle in the preceding example, and the other two paths are straight line segments. The integral along the horizontal interval on the x -axis is 0, because $xy = 0$ there. The integral along the vertical interval on the y -axis is 0, because $dx = 0$ there. (To see this, either parametrize the line segment explicitly, or go back to the definition (1.1) and observe that each of the x_j 's is 0.) Using the result of the preceding example, we find that the value of the integral around ∂D is $-\frac{1}{3}$.

A very useful tool for evaluating line integrals is provided by Green's theorem, which converts a line integral around the boundary of a domain to an area integral over the domain.

Green's Theorem. *Let D be a bounded domain in the plane whose boundary ∂D consists of a finite number of disjoint piecewise smooth closed curves. Let P and Q be continuously differentiable functions on $D \cup \partial D$. Then*

$$(1.4) \quad \int_{\partial D} P \, dx + Q \, dy = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y} \right) dx \, dy.$$



Example. We again evaluate $\int_{\partial D} xy \, dx$, where D is the quarter-disk in the first quadrant, this time using Green's theorem. In this case, $P(x, y) = xy$ and $Q(x, y) = 0$, so (1.4) becomes

$$\begin{aligned} \int_{\partial D} xy \, dx &= - \iint_D x \, dx \, dy = - \iint r \cos \theta \, r \, dr \, d\theta \\ &= - \int_0^{\pi/2} \cos \theta \, d\theta \int_0^1 r^2 \, dr = -(1)\left(\frac{1}{3}\right), \end{aligned}$$

as before.

Since Green's theorem is of fundamental importance, we provide a sketch of the ideas behind the derivation of the formula (1.4). One basic idea is to cut the domain into little curvilinear triangular pieces and treat each piece separately. Another is to reduce the double integral over a triangle

to a line integral by applying the fundamental theorem of calculus in one variable with the other variable as a parameter. For convenience, we break the proof-sketch into three steps.

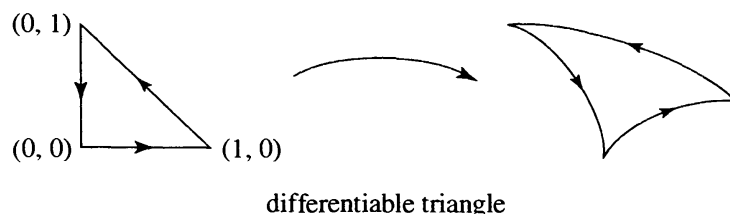
The first step is to establish the formula (1.4) for the triangle T with vertices at $(0, 0)$, $(1, 0)$, and $(0, 1)$. We must establish the two identities

$$\int_{\partial T} P dx = - \iint_T \frac{\partial P}{\partial y} dx dy, \quad \int_{\partial T} Q dy = \iint_T \frac{\partial Q}{\partial x} dx dy.$$

Consider just the first identity here. We represent the double integral as an iterated integral and use the fundamental theorem of calculus, to obtain

$$\iint_T \frac{\partial P}{\partial y} dx dy = \int_0^1 \left[\int_0^{1-x} \frac{\partial P}{\partial y} dy \right] dx = \int_0^1 P(x, 1-x) dx - \int_0^1 P(x, 0) dx.$$

The sum on the right we recognize as $-\int_{\partial T} P dx$, after we parametrize separately the three sides of ∂T . Indeed, the line integral of $P dx$ along the vertical edge of T is 0, since $dx = 0$ there; the line integral of $P dx$ along the bottom edge of T is $\int_0^1 P(x, 0) dx$; and the line integral of $P dx$ back along the hypotenuse of T is $-\int_0^1 P(x, 1-x) dx$, where we have used the parameterization $y = 1 - x$.



The second step of the proof is to establish the formula for any domain D that can be obtained from the triangle T by a change of variables. (See Exercise 7.)

The final step in the proof, for an arbitrary domain D , involves triangulating D , that is, cutting D into small triangular pieces, each of which can be obtained from the triangle T by a change of variables. Green's theorem is applied to each triangular piece, and the results are added. The sum of the area integrals over the triangular pieces is the area integral over D . The boundary integrals over the sides of the triangular pieces inside D cancel in pairs, since each curvilinear triangle side is traversed twice, once in each direction, and the opposing directions cancel. The boundary integrals over the curvilinear triangle sides in ∂D add up to the integral over ∂D .

